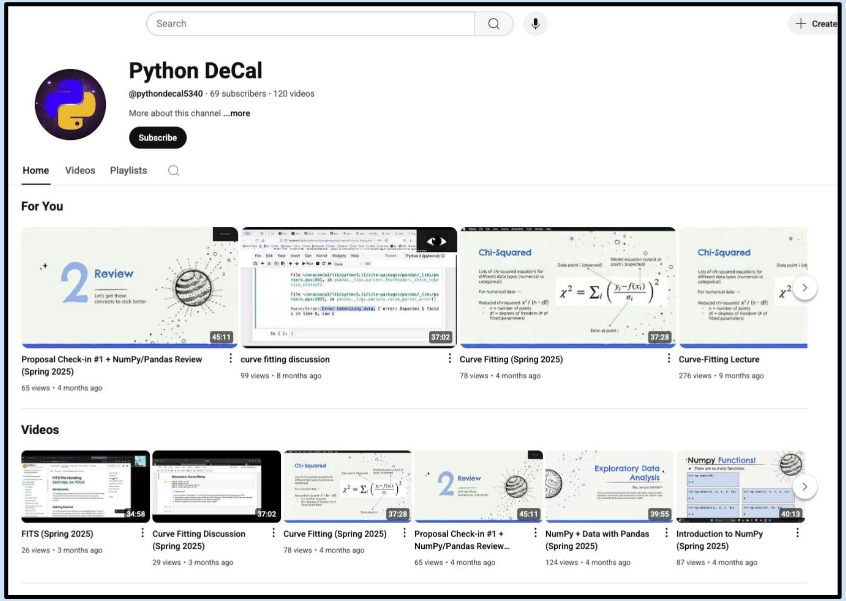
Welcome to the Python Decal!

Lecture: GitHub

[Class 3] September 8th, 2025

LECTURES WILL BE RECORDED



This class will be **recorded** and posted to our YouTube Channel!

Subscribe :)

Class Playlist on Spotify!

Scan the QR if you would like to see our class playlist!

We will try and play it before each class, remind us if we forget...

Also posted on bCourses

Overview for Today!

1. Announcements
2. Warm-Up
3. Demo
    a. Initialize a local repository
    b. Saving Your Work
    c. Initialize a remote repository
    d. Connecting with GitHub
4. Key Takeaways (because y'all, GitHub is confusing)
    a. Practice Question
5. Lecture Check

01

Announcements

:) :) :) :) :) :) :) :) :) :) :) :)

Announcements

Welcome to the first real week of lecture + discussion!

This will be the typical format as we move throughout the semester.

This week we will be focusing on GitHub.

Announcements

- HW #1 is due this Wednesday, Feb. 4th at 11:59 PM
- HW #1 late due date is Feb. 18th at 11:59 PM
  - Next, next Wednesday

- HW #2 was released today Monday, Feb. 2nd
- HW #2 is due next Wednesday, Feb. 11th at 11:59 PM
- HW #2 late due date is Feb. 25th at 11:59 PM

- Office hours are being finalized this week!

02

Warm-Up

Let’s do a little review

warm-up questions

Please answer these on a piece of paper:

1) Define the following commands:
   a) pwd,
   b) ls,
   c) cd,
   d) mkdir,
   e) nano.
2) Define the following data types: string, integer, float.
3) What are the <u>two</u> ways to run python code on the terminal?
   -

Warm-up Questions Answers

1) pwd = print current working directory
   ls = list files and folders in current working directory
   cd = change directories
   mkdir = make a new directory
   nano = opens a text file editor
2) string = text/language, integer = whole numbers, floats = fractional numbers
3) Call: python3 <script>.py or call python3 and then type python code directly on the command line

03

Demo

GitHub: version control

Prelude to Today's Demo

Disclaimer:

Working with Git and GitHub can be hard!

It is difficult to learn and <u>not</u> intuitive at the beginning.

Best advice:

Stick with it and keep practicing! I promise it will make sense

Resources:

We have created the <u>The GGG</u> (Git and GitHub Guide),

feel free to look it over if you need help!

Background Information

To save our work we need to follow 3 important steps with git:

- Add our work to the staging area
- Commit (save) our work locally
- Push (save) our work remotely

Local = on your computer

Remote = on the internet

Background Information

We will be covering A LOT of detail today, some things are super important for you to remember

- The detail is for your future reference
- Check out The GGG if you need a reference
- It's good to know the ins and outs of Git for debugging

At the end of the demo, I will tell you what the key takeaways from today should be

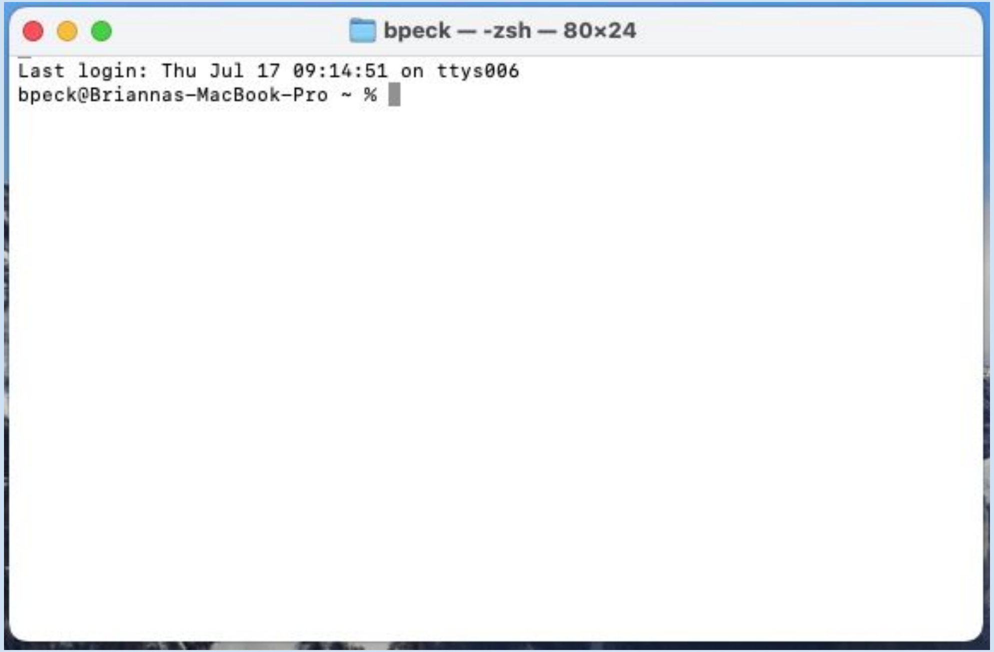
Initialize a <u>local</u> repository

Background Information

The steps below are <u>NOT</u> ones you need to do every single time you want to save your work. These only need to happen once.

Next class, we will spend time practicing just saving our work in general

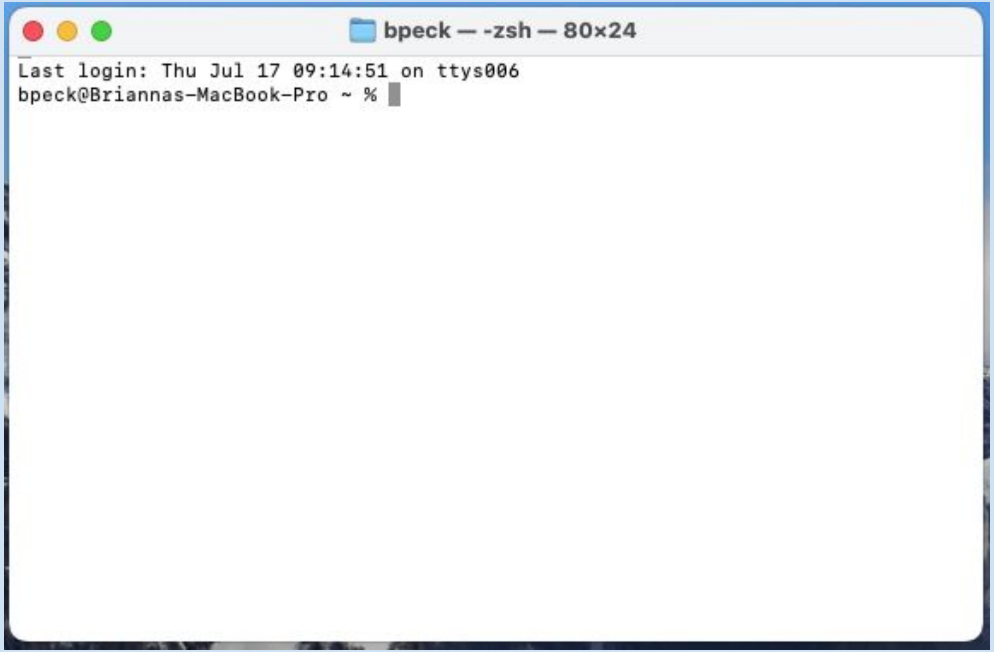
OPEN THE TERMINAL



Open your terminal

Windows Users: Make sure you are using GitBash as your shell and NOT PowerShell!

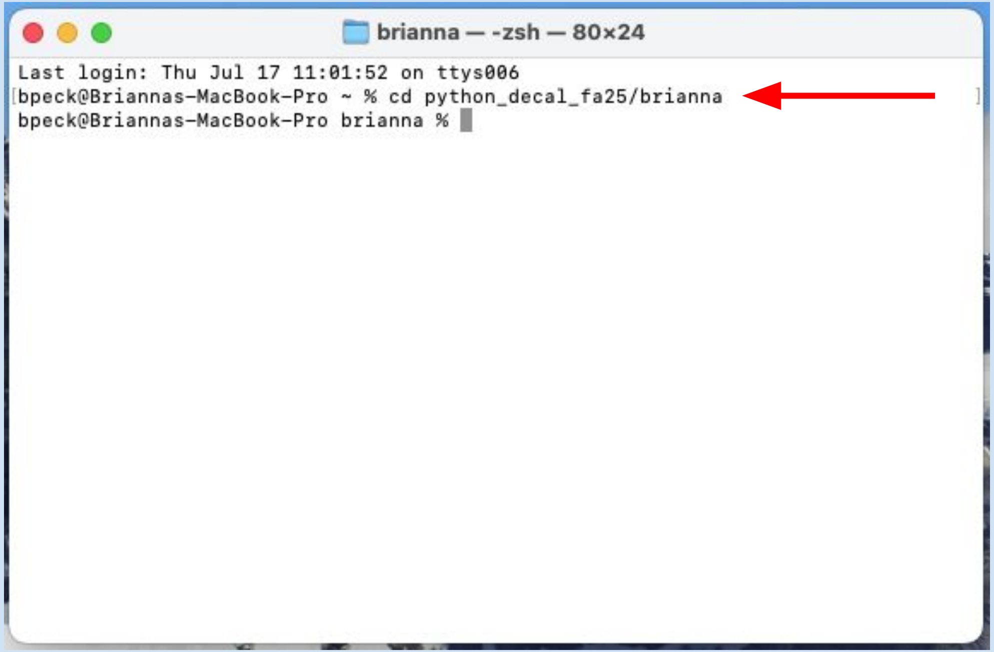
OPEN THE TERMINAL



Recall: last class (and as a part of homework 1), you made a folder called <u>yourname</u>

Please make sure you have followed those directions before continuing

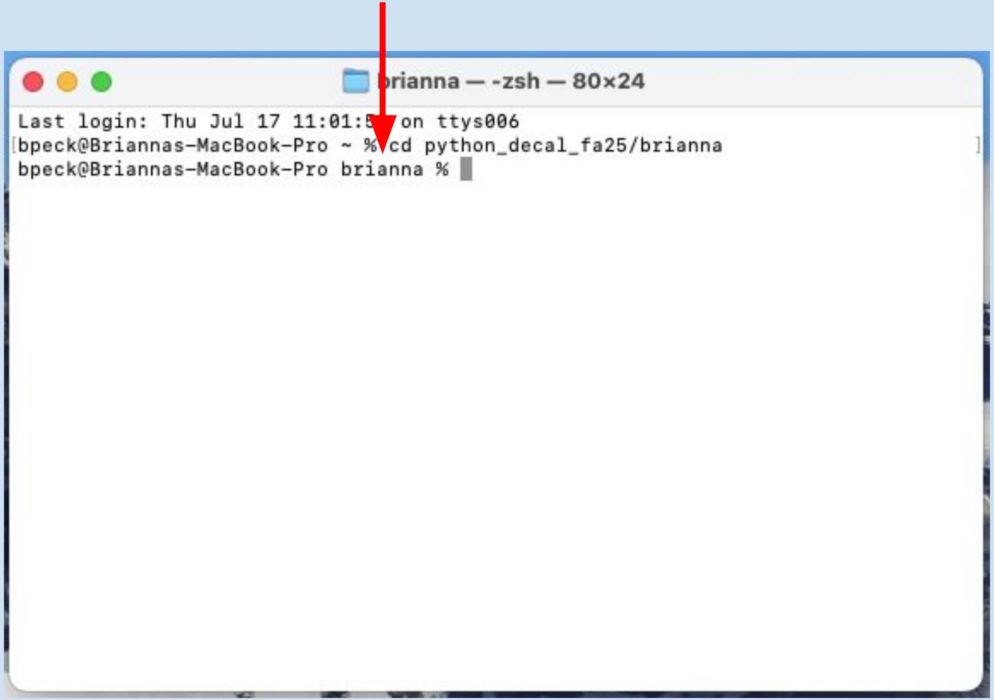
Change Directories



Move into your <u>yourname</u> folder by writing a file path

My name is Brianna, so my folder is called <u>brianna</u> :)

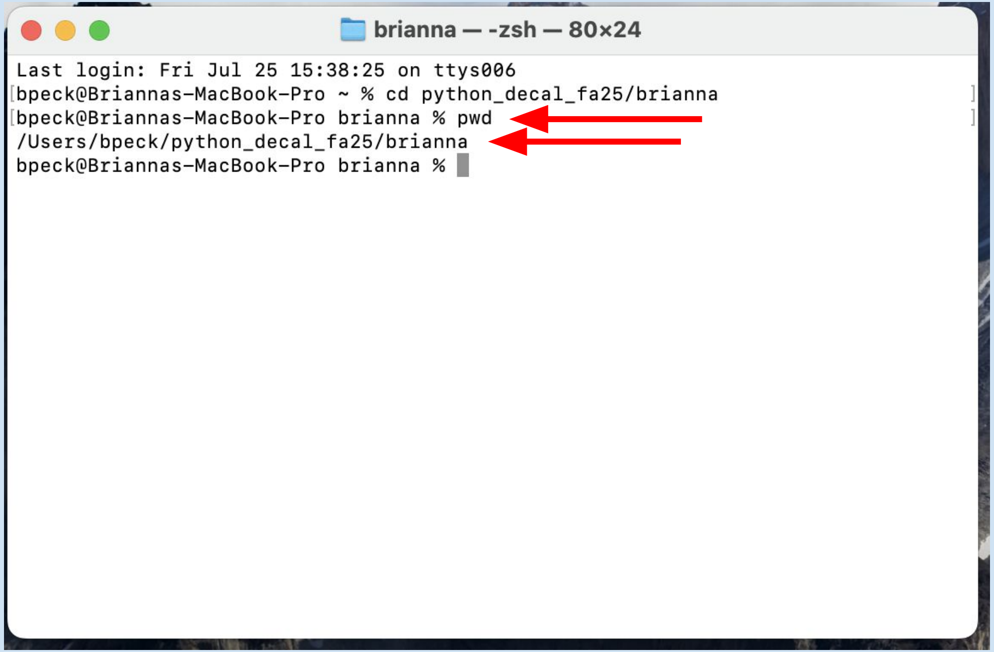
Change Directories



Your directory may tell you your current working directory

If it doesn't, no worries

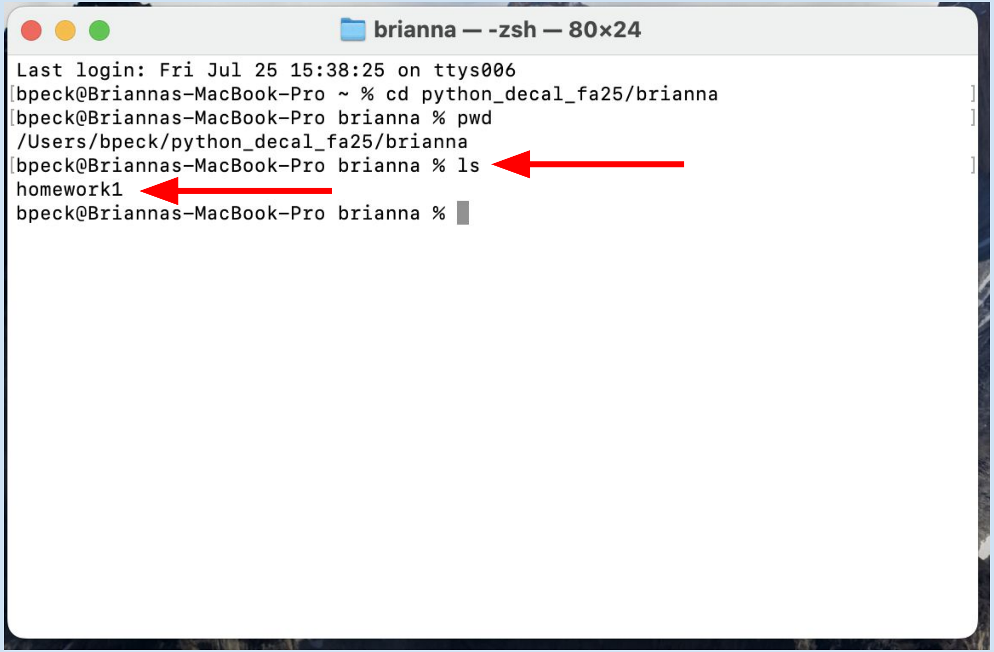
Print Working Directory



But just to be safe, let's call pwd which will print our current working directory

We are in the folder yourname, which is in the folder python_decal_fa25

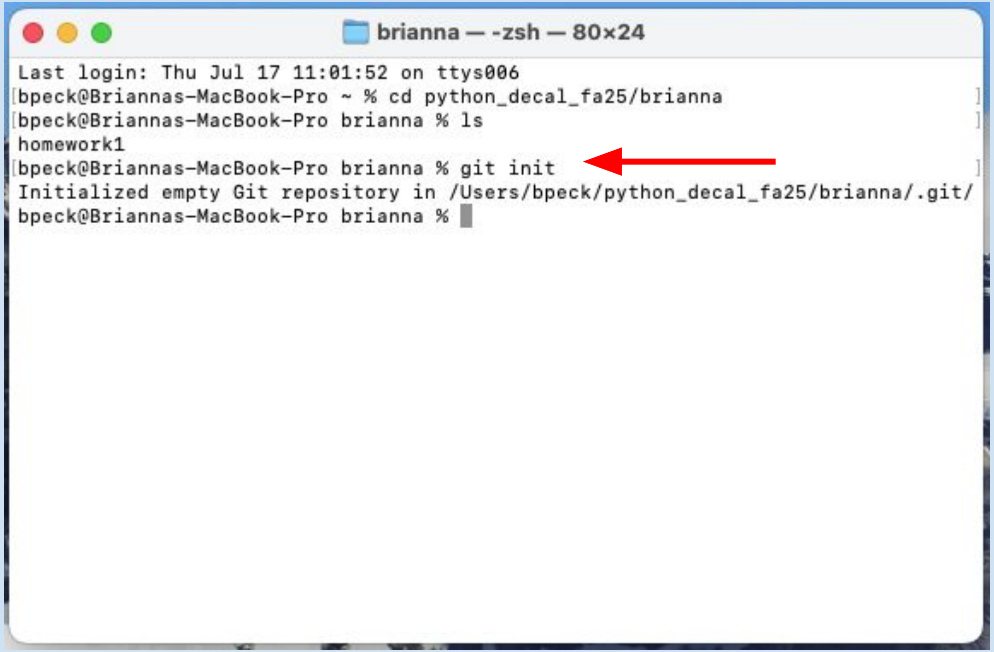
LIST THE CONTENTS



Next, let's list the contents of the directory

If you have been working on homework 1, then you should have a folder called homework1/

Initialize a repository

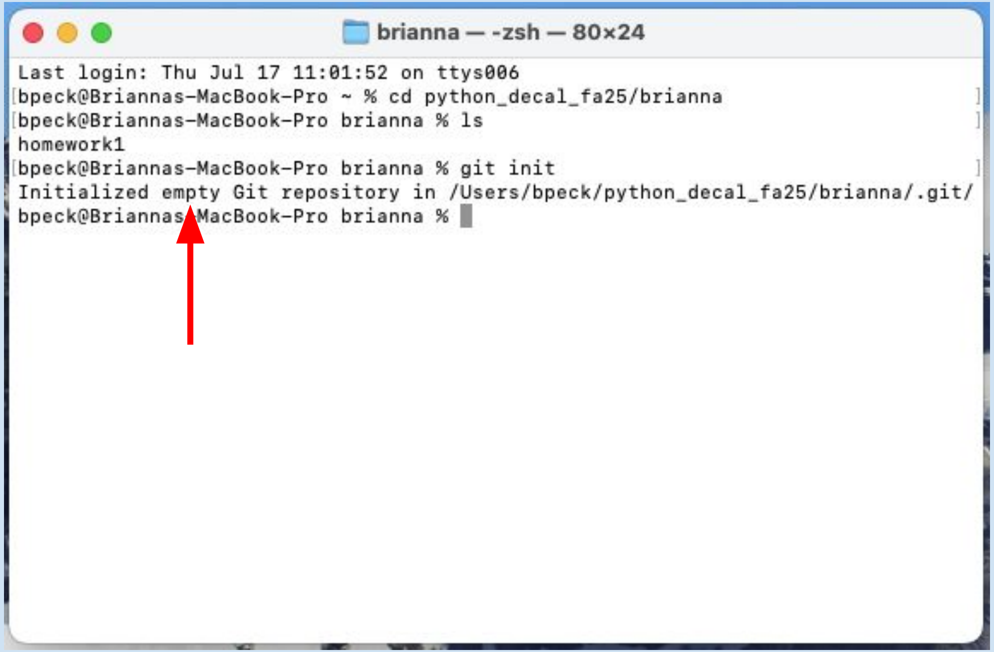


To create a repository out of our directory, type:

<u>git init</u>

Repository = a directory that has version control abilities!

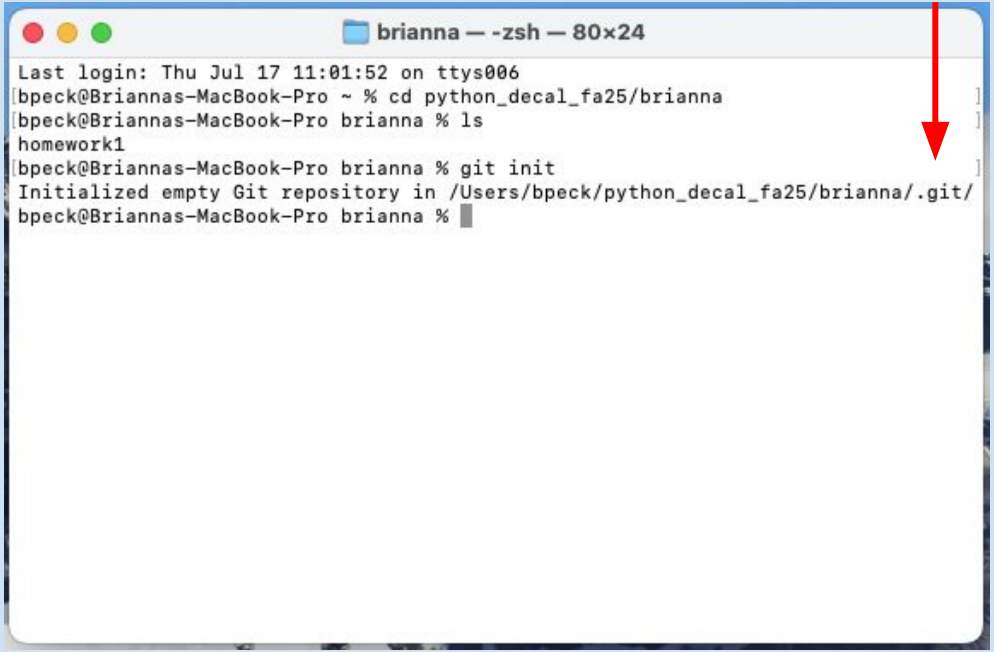
Initialize a repository



When we look at the message that popped up, we see that our repository is "empty"

That means we don't have any saved changes yet, we need to <u>manually</u> save our changes every single time

HIDDEN GIT FOLDER

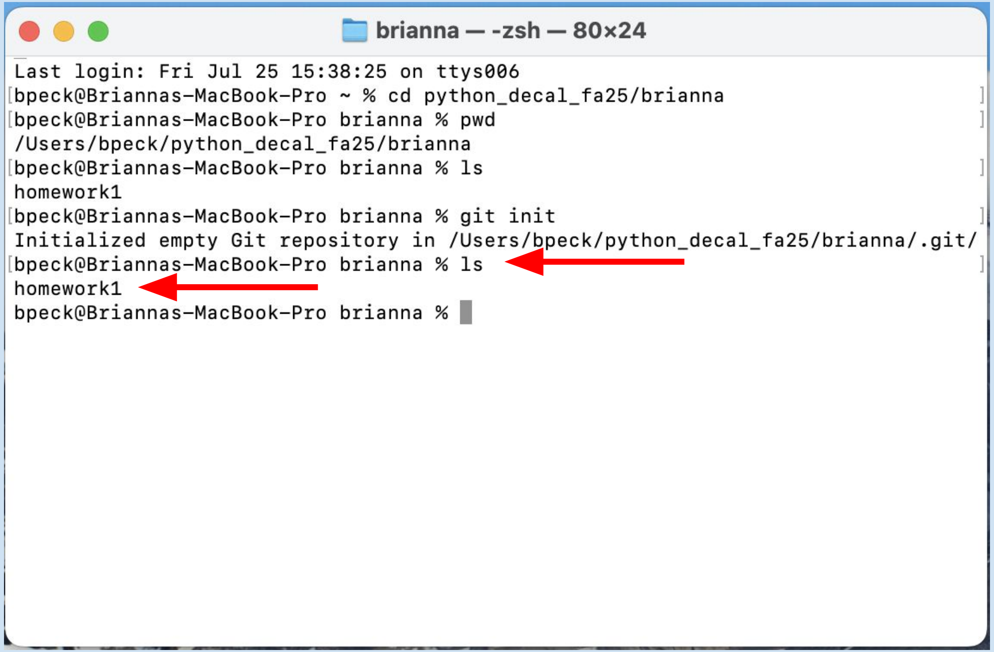


When we look at the filepath, we see this new folder: .git/

This is actually a hidden folder and won't appear if we just call ls

It holds the information for our version control

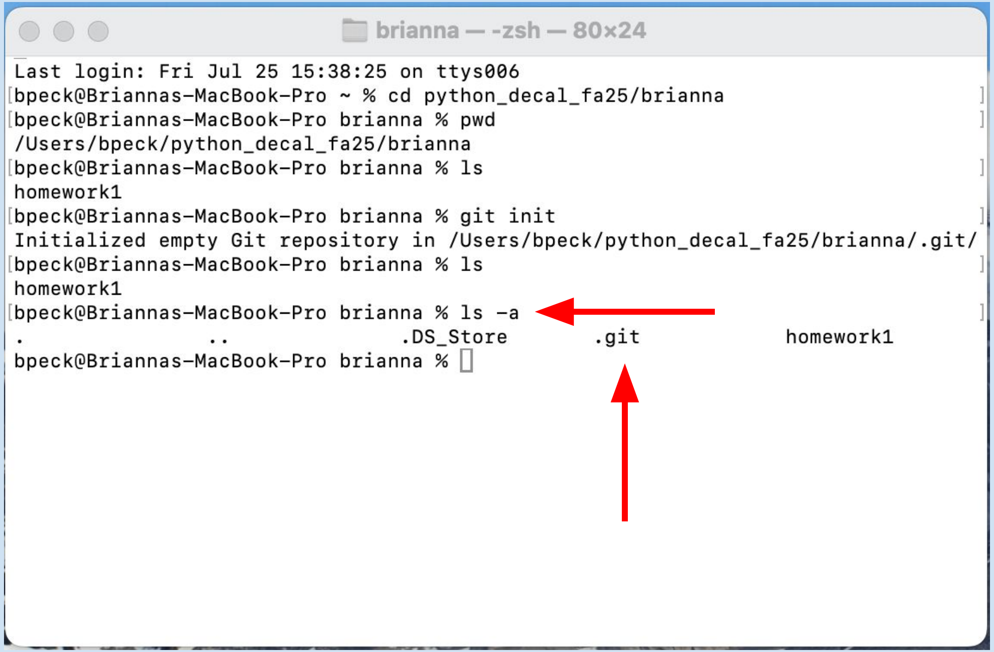
LIST NON-HIDDEN CONTENTS



So when we call ls we don't see that .git/ folder, we only see the original homework1/

But if we call ls -a, which shows hidden files, we will see it

LIST HIDDEN CONTENTS



The hidden .git/ folder appeared, it will store information in the background

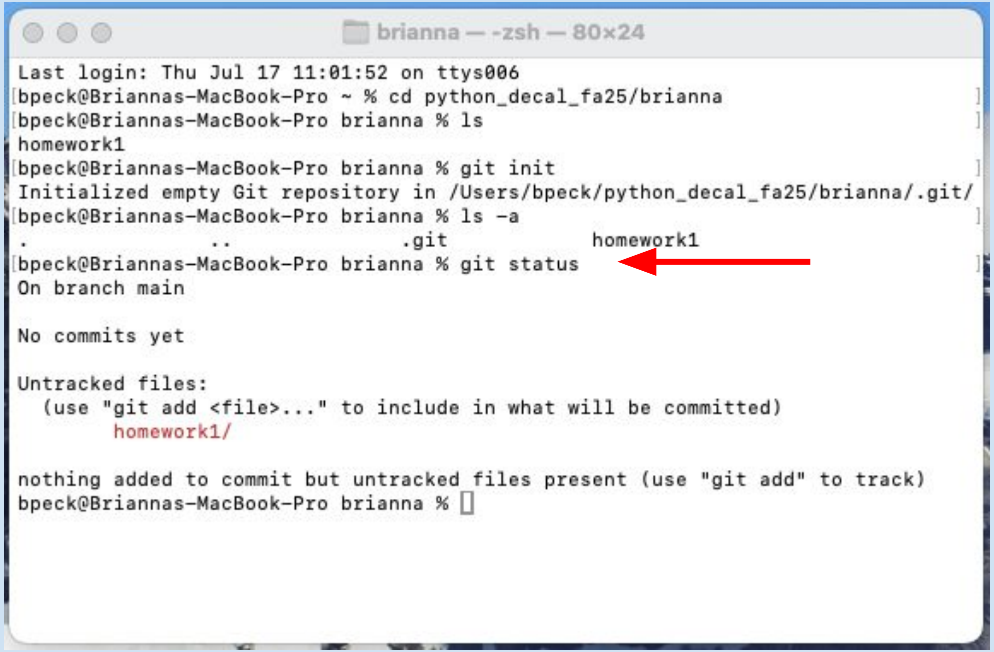
The rest mean:

. = current directory

.. = parent directory

.DS_Store = created by the mac Finder, irrelevant

Call git status

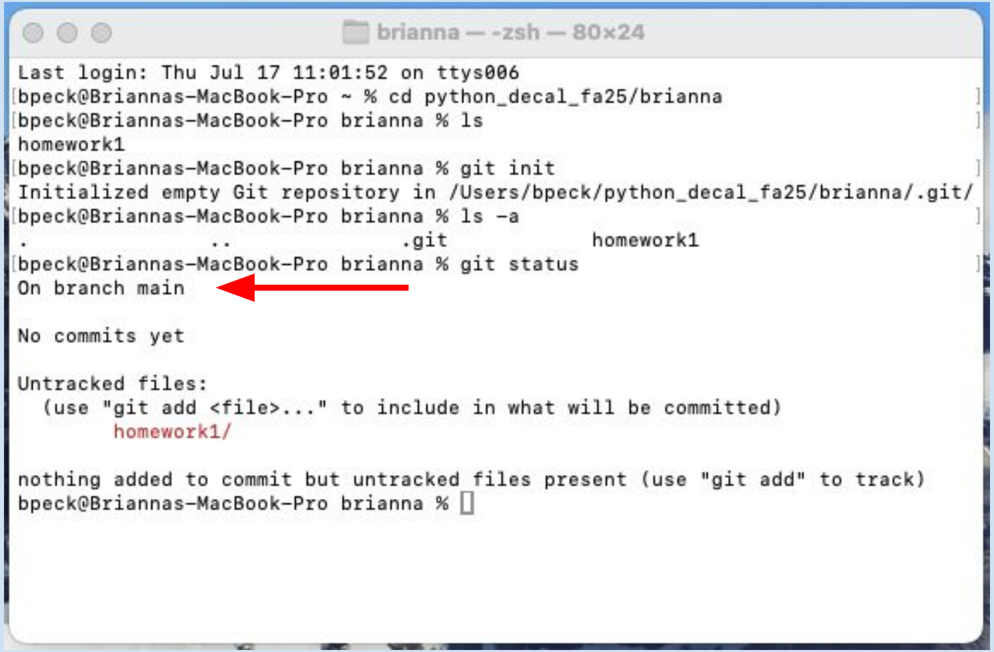


Let's call a command that is really helpful for <u>debugging</u>

<u>git status</u>

It is a command that tells you the "status" of your repository

Main is our Branch

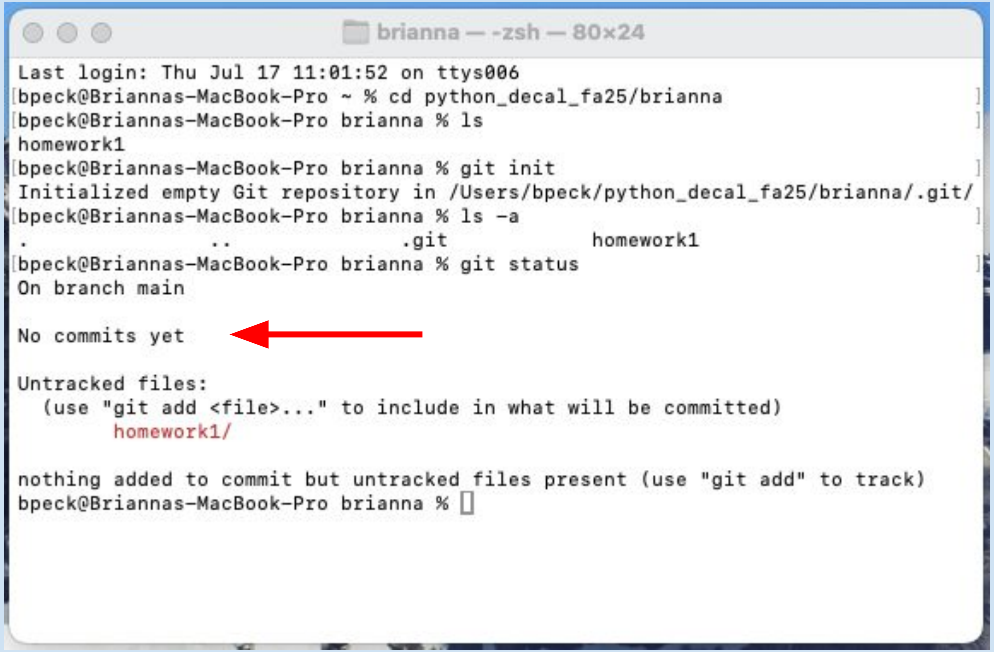


Let's work through line-by-line what it has told us

<u>On branch main</u>

We haven't covered branches, but this tells us the name of the branch we are on, so <u>main</u>

an empty repository



```
brianna — -zsh — 80×24
Last login: Thu Jul 17 11:01:52 on ttys006
[bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
[bpeck@Briannas-MacBook-Pro brianna % ls
homework1
[bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
[bpeck@Briannas-MacBook-Pro brianna % ls -a
.               ..              .git            homework1
[bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % 
```

<u>No commits yet</u>

This means we have saved <u>no changes</u> yet, we are NOT performing version control yet

So our repository is still acting like a directory

GIT sees changes

```
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
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On branch main

No commits yet

Untracked files:
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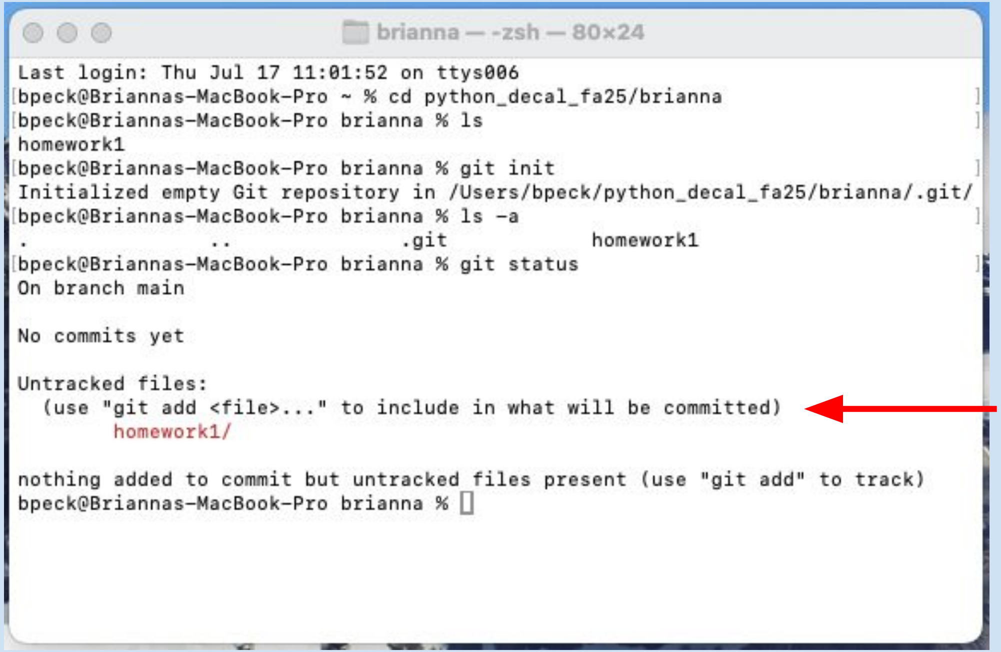
nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % 
```

Untracked files:

This means git sees that there are files in the directory that we haven't backed up yet

Secret lecture code: save

GIT TELLS US WHAT IT SEES



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On branch main

No commits yet

Untracked files:
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```

(use "git add <file>..." ...)

This line, is git telling us what to do to save our work

We will call git add shortly

GIT TELLS US HOW TO SAVE WORK

```
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On branch main

No commits yet

Untracked files:
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bpeck@Briannas-MacBook-Pro brianna % 
```

<u>homework1/</u>

This is git telling us that we have a folder called homework1, that has NOT been saved

git does NOT save work automatically

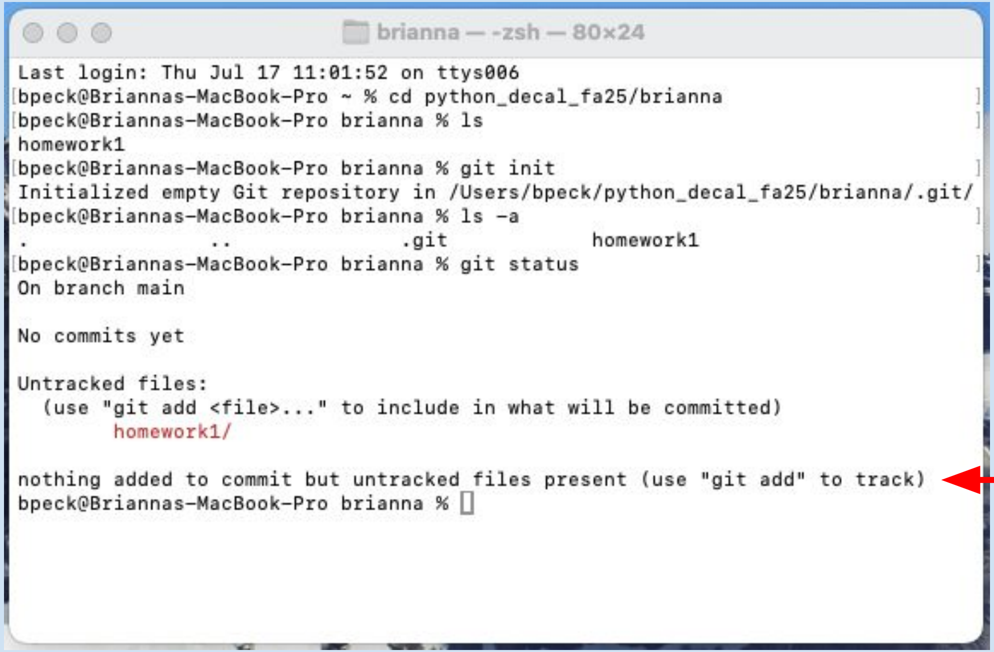
saving your code

Background Information

As we go through these steps, your terminal might give you slightly different responses

- Especially if you have NOT started homework 1
- This lecture assumes that you have started creating your *homework1.py* Python file

GIT TELLS US HOW TO SAVE WORK



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On branch main

No commits yet

Untracked files:
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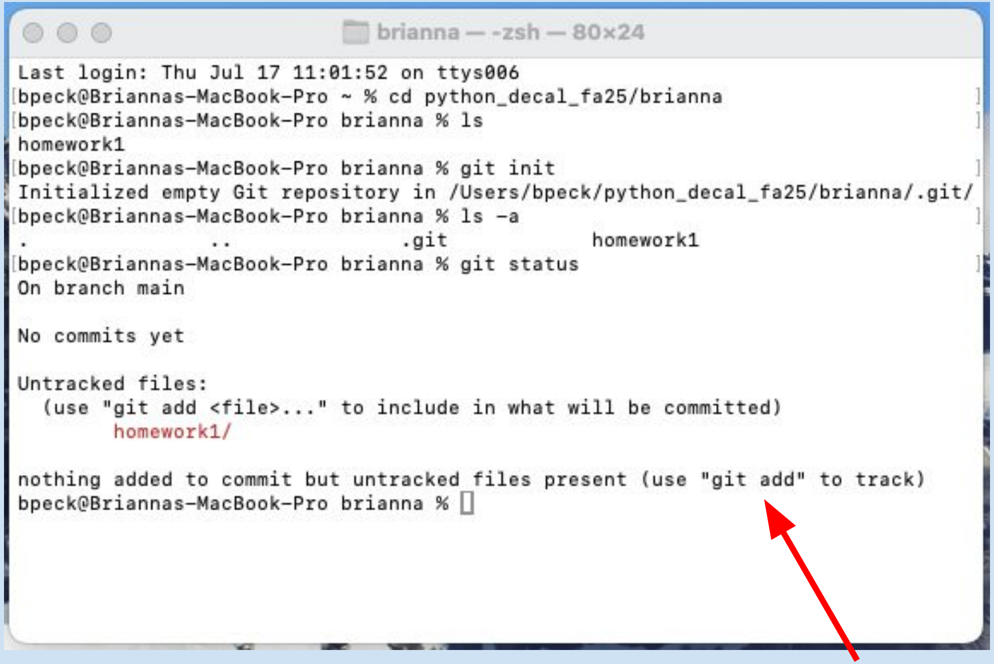
        homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

<u>nothing added to commit...</u>

This message is once again telling us we need to call: <u>git add</u>, to save our work

GIT TELLS US HOW TO SAVE WORK



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        homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

There are two main ways to call <u>git add</u>

1) <u>git add .</u>

This saves ALL your files

2) <u>git add <file></u>

This saves only the files you specify

ADD TO STAGING AREA

brianna — -zsh — 80×24

```
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
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.               ..              .git            homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        homework1/

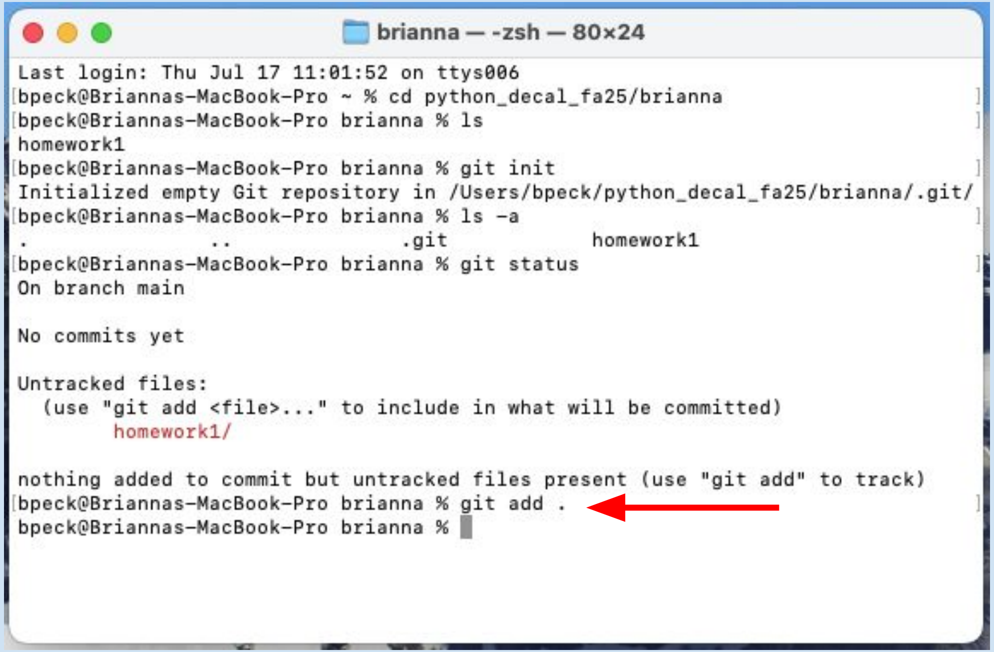
nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
```

For now, we want to save everything in this repository so let's call git add .

Now git will search through all of yourname/ and looks for files to save

But we are not done yet, there are more steps

ADDED TO STAGING AREA

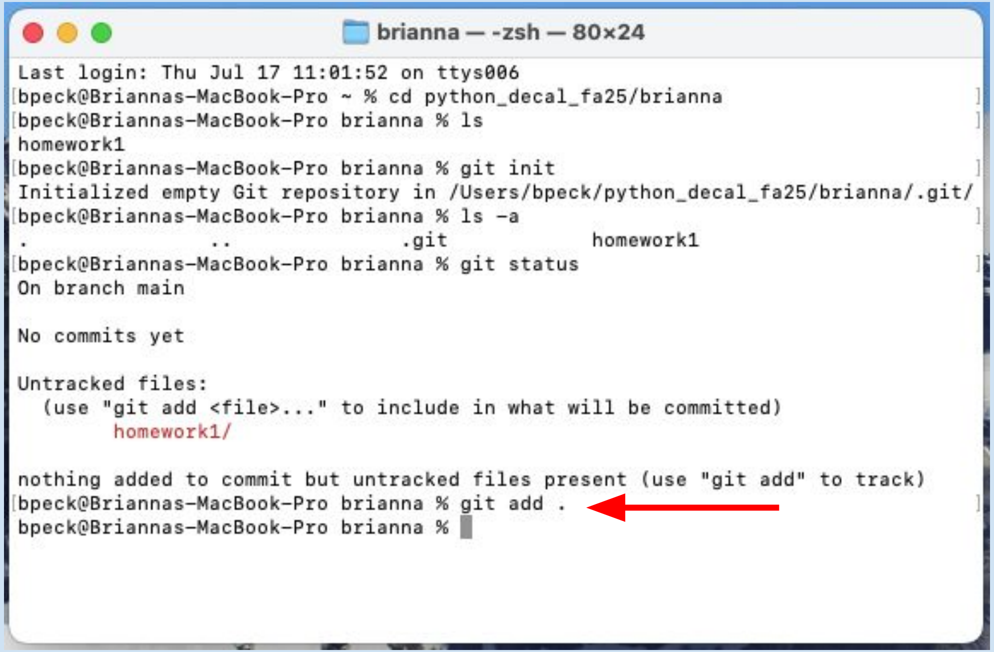


Once you have called:

<u>git add .</u>

Nothing will appear! Git is now “tracking” these files and they are in a “staging” area

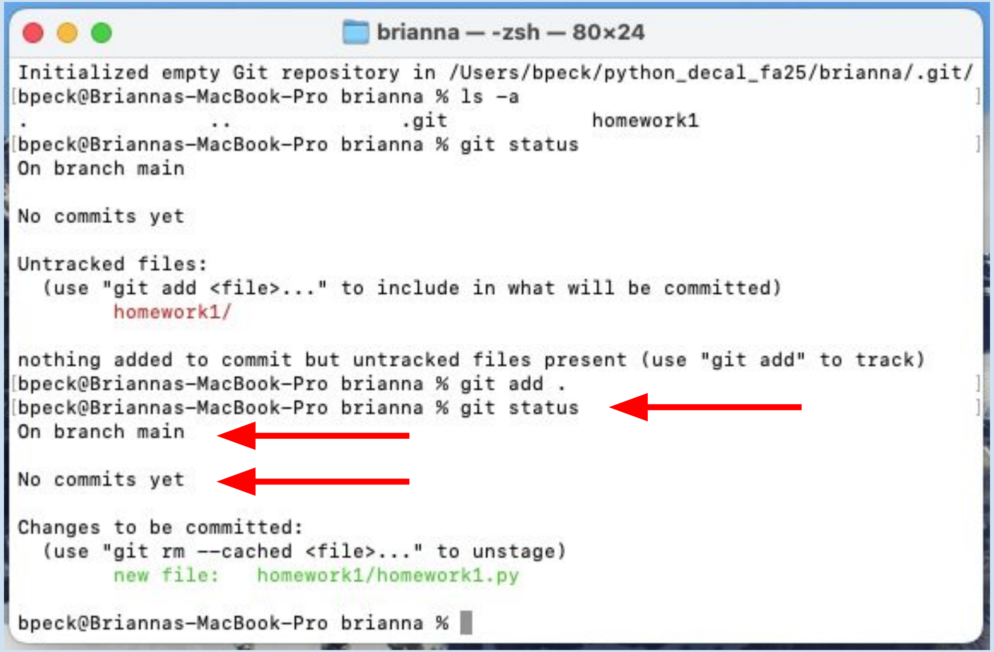
Added to staging area



<u>Tracking</u> = Git knows about the existence of a file and is keep a history of its changes

<u>Staging Area</u> (rough draft zone) = a place to make last-minute changes before committing files to your saved history

GIT STATUS GIVES US AN UPDATE



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On branch main

No commits yet

Untracked files:
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nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   homework1/homework1.py

bpeck@Briannas-MacBook-Pro brianna %
```

Let's call:

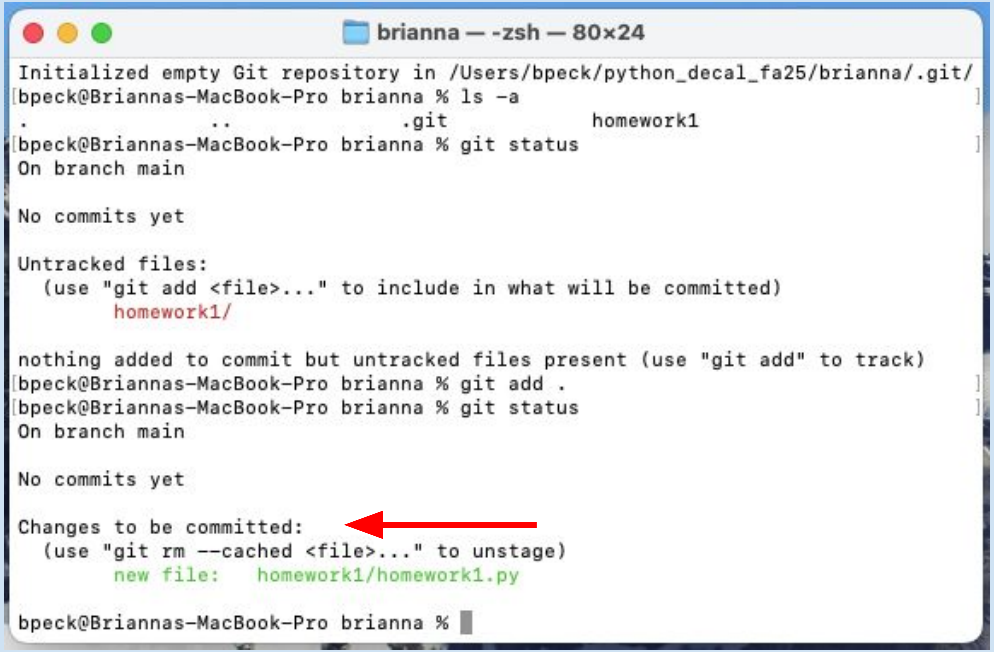
git status again

We see these lines again:

- On branch main
- No commits yet

Meaning, we haven't changed branches or completely saved our files

stuff in staging area



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On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   homework1/homework1.py

bpeck@Briannas-MacBook-Pro brianna % 
```

<u>Changes to be committed:</u>

This means we have <u>added</u> files to our staging area (rough draft zone)

If we want to <u>unadd</u> files, this would be the time to do it

HOW TO DELETE FROM AREA

```
brianna — -zsh — 80×24
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.               ..              .git            homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
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bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   homework1/homework1.py

bpeck@Briannas-MacBook-Pro brianna %
```

(use “git rm –cached <file...

For example, if we added a multiple versions of a file, and only want to keep one of them

We could call git rm to remove them from the staging area

How to Delete from Area

```
brianna — -zsh — 80×24
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.               ..              .git            homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
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On branch main

No commits yet

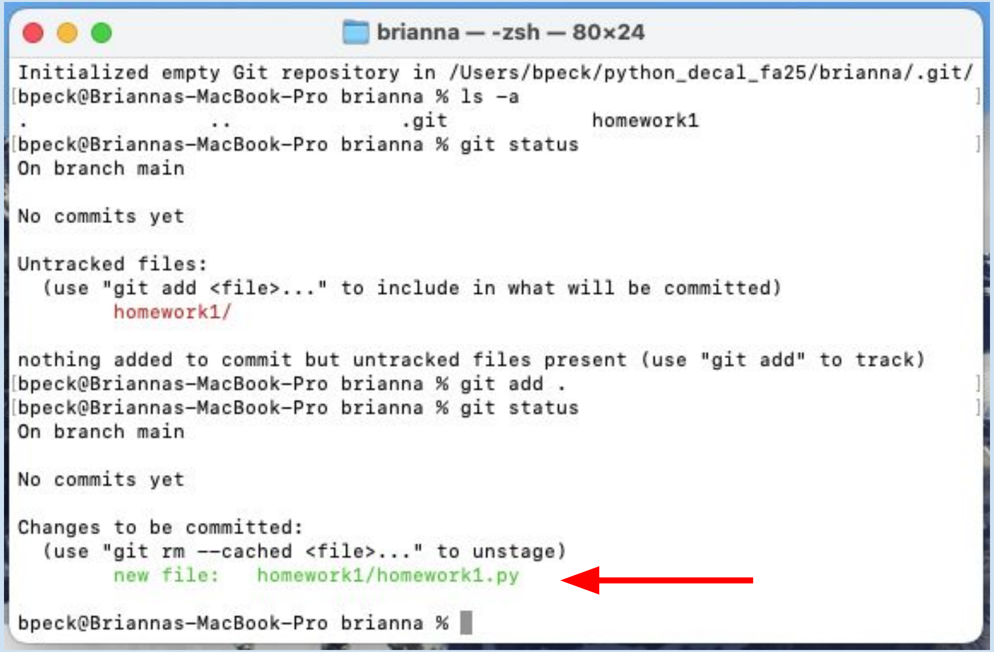
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   homework1/homework1.py

bpeck@Briannas-MacBook-Pro brianna % 
```

But we want to just save the changes on ALL of our files right now

So we won't be calling git rm

What File Is In The Area

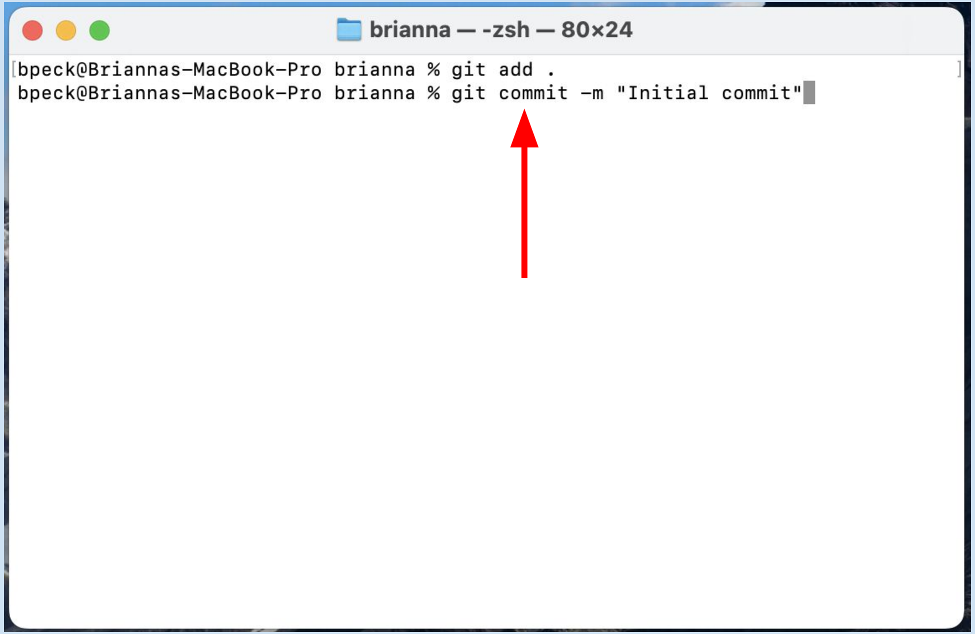


<u>new file: homework1/hom...</u>

This line tells us that a new file has been <u>added</u> to the <u>staging area</u>

Git is awaiting us to fully save its progress

saving work to local comp.



The next step in the process of saving our work is to call <u>git commit</u>

The full command is:
<u>git commit -m “message”</u>

For the message you can write something like:
<u>Saving homework1/ folder</u>

saving work to local comp.

```
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna % 
```

Git will give you a really long message in return, let's work through a bit of the message <u>line-by-line</u> again

The Commit Message

```
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```

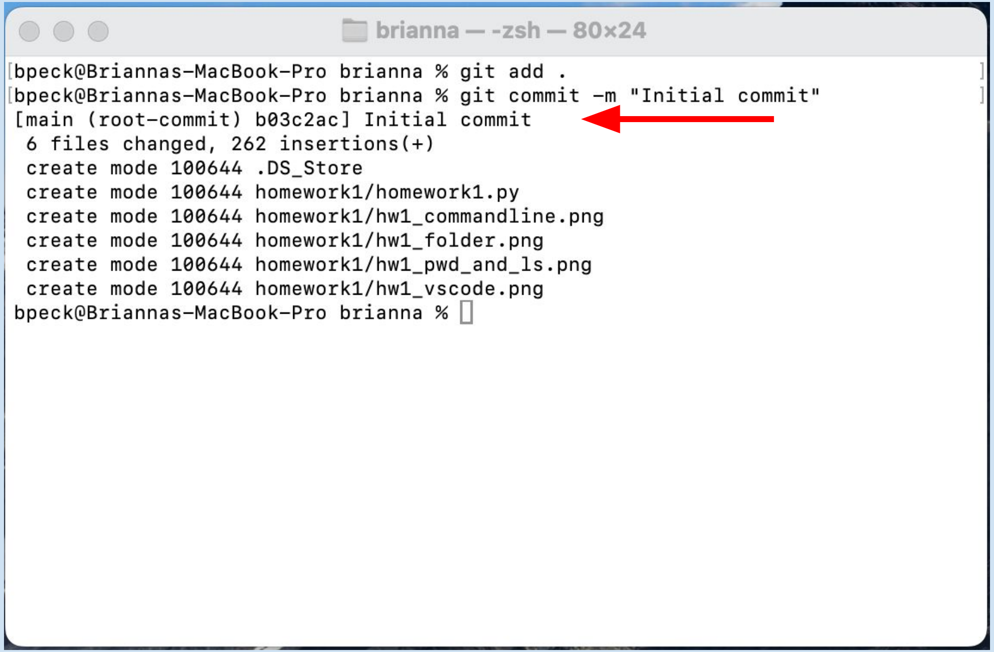
<u>[main (root-commit) b03c2ac]</u>

<u>main</u> = the branch you are on

<u>(root-commit)</u> = the command you just called created the first commit of the repository

<u>b03c2ac</u> = the commit number (i.e., ID #) of the version you just saved

THE COMMIT MESSAGE



Initial Commit

This is the message associated with the version you just saved to your computer

The Commit Message

```
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```

This is git telling us that we have changed six files with 262 line insertions

- So, six files are being saved for the first time
- In all of those files, 262 lines of code have been saved

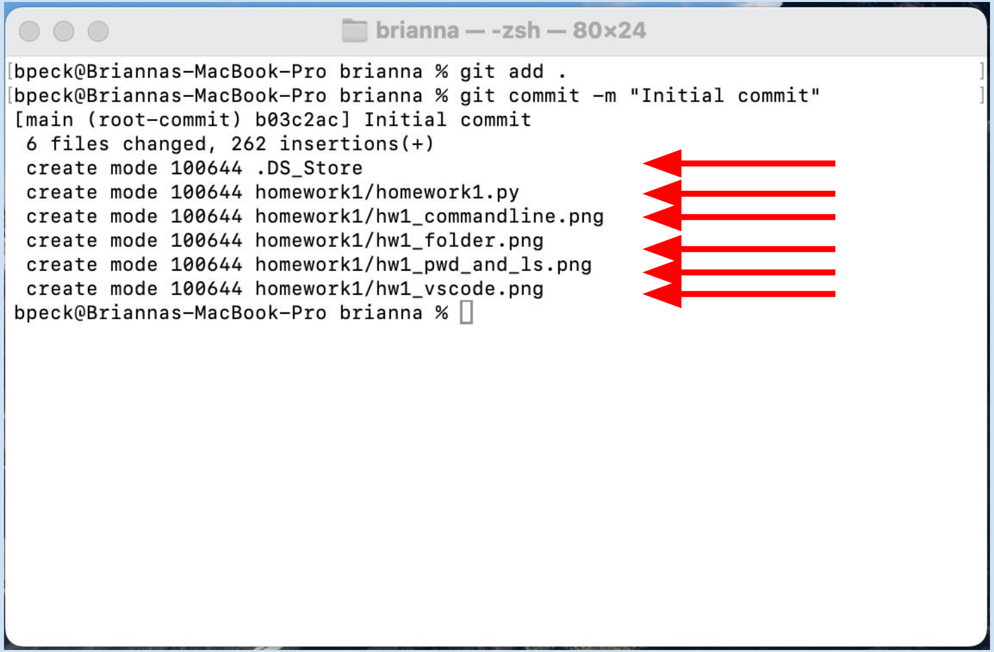
The Commit Message

```
brianna — -zsh — 80×24
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```

Since this is the first time we are saving changes, git has nothing to compare the changes against

So this line just tells us what is currently in our repository prior to saving our work

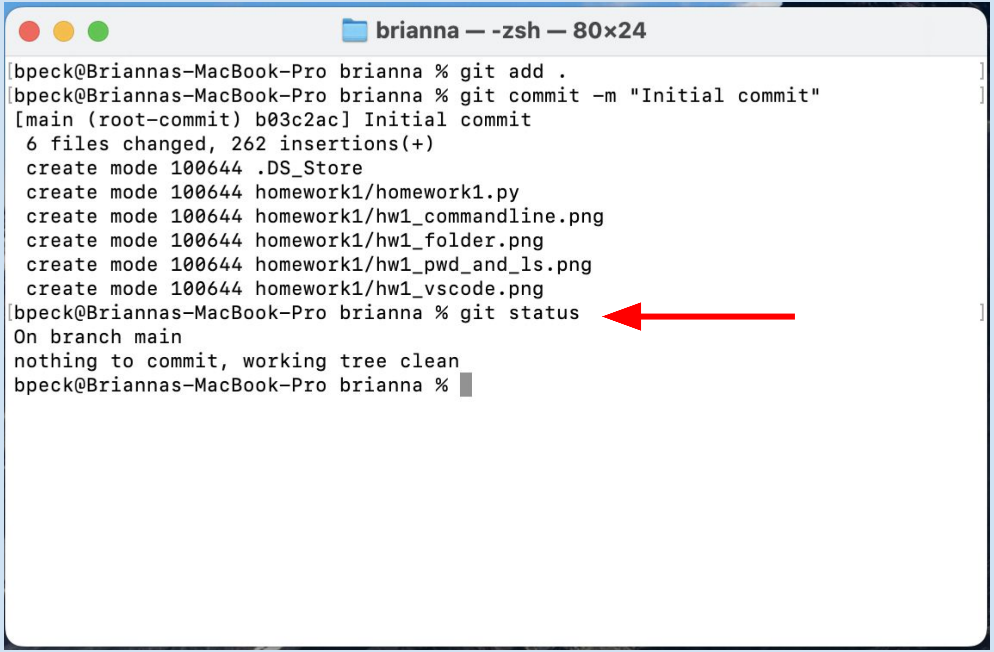
THE COMMIT MESSAGE



All of these lines means <u>different files</u> that are being saved for the first time in our version history

If you have completed <u>homework 1</u> this is what your repository would look like

Calling Git Status for Update

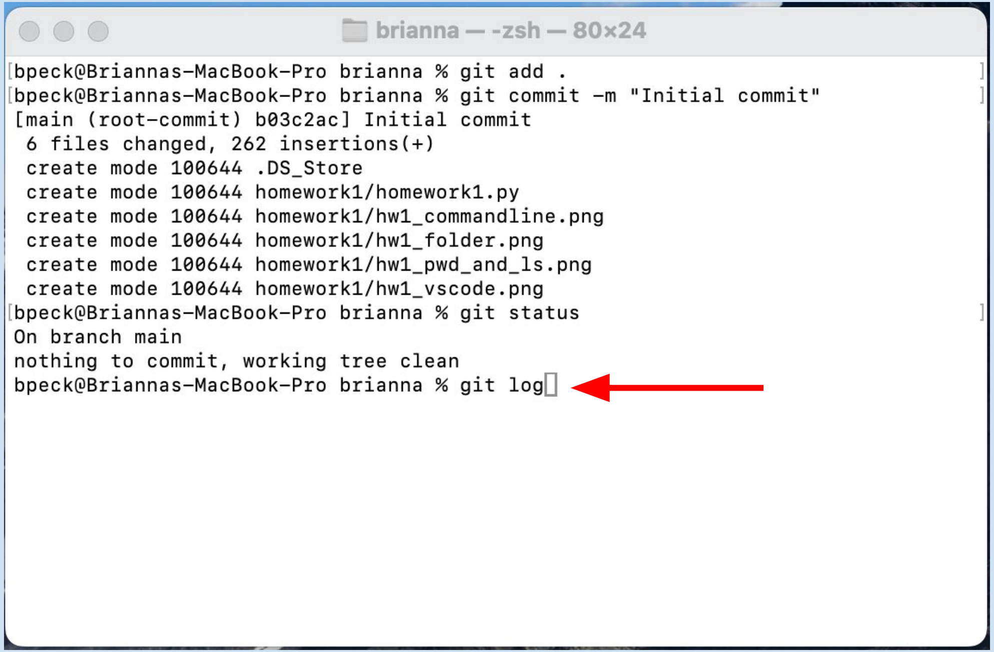


Let's call git status again

We see the original:
- On branch main

But now we see:
- Nothing to commit, working tree clean
- So Git has properly saved all of our work!

Call git log for history



The command git log is also pretty helpful, it tells us the actual history of our folder

Call GIT LOG for History

```
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git log
commit b03c2ac3285a925df67386af3c7896ed6b74fd53 (HEAD -> main)
Author: Brianna Peck <bpeck114@berkeley.edu>
Date:   Sat Jul 26 08:17:24 2025 -1000

    Initial commit
bpeck@Briannas-MacBook-Pro brianna %
```

There is our commit number again, but this time the <u>full commit number</u>

<u>(HEAD -> main):</u>

<u>HEAD</u> = the current commit number you are using

<u>-> main</u> = you are using the latest commit on your main branch

Call Git Log for History

```
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git log
commit b03c2ac3285a925df67386af3c7896ed6b74fd53 (HEAD -> main)
Author: Brianna Peck <bpeck114@berkeley.edu>
Date:   Sat Jul 26 08:17:24 2025 -1000

    Initial commit
bpeck@Briannas-MacBook-Pro brianna %
```

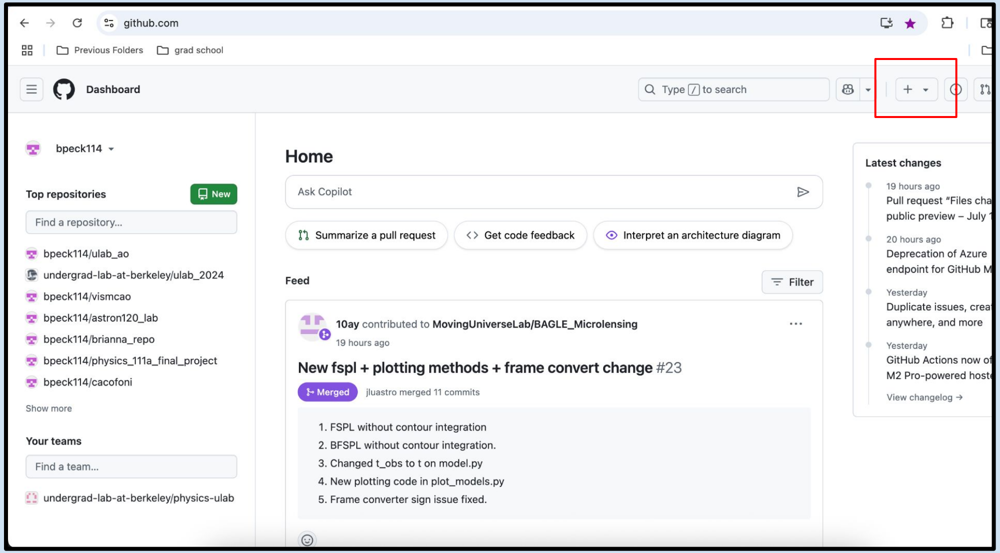
The person who made the commit and their associated email address

The date and time the commit was saved

The message of the commit

Initialize a <u>remote</u> repository

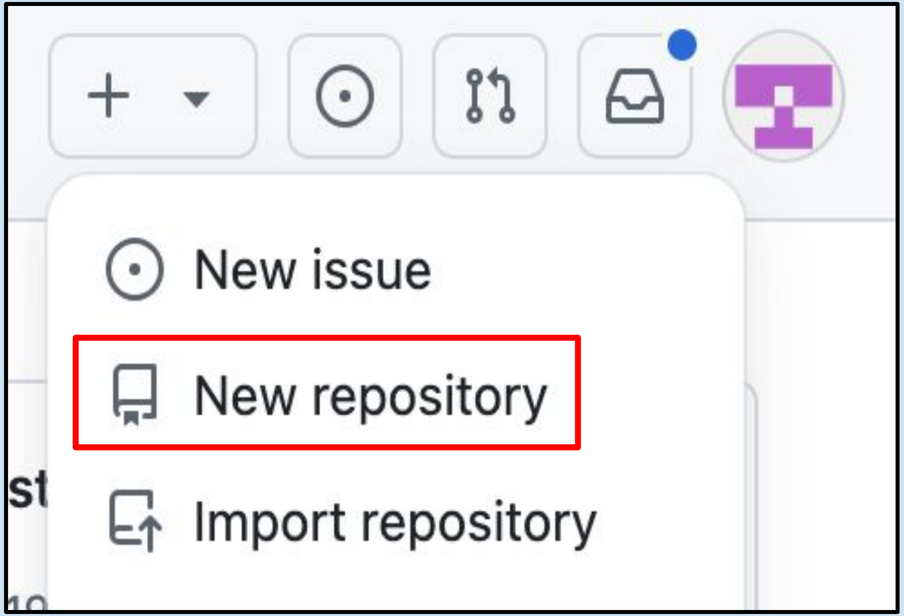
Open GitHub



On a web browser, open up GitHub. Sign in with your @berkeley.edu email address.

On the top right-hand corner, you will see a box with a plus sign. Click on that plus sign.

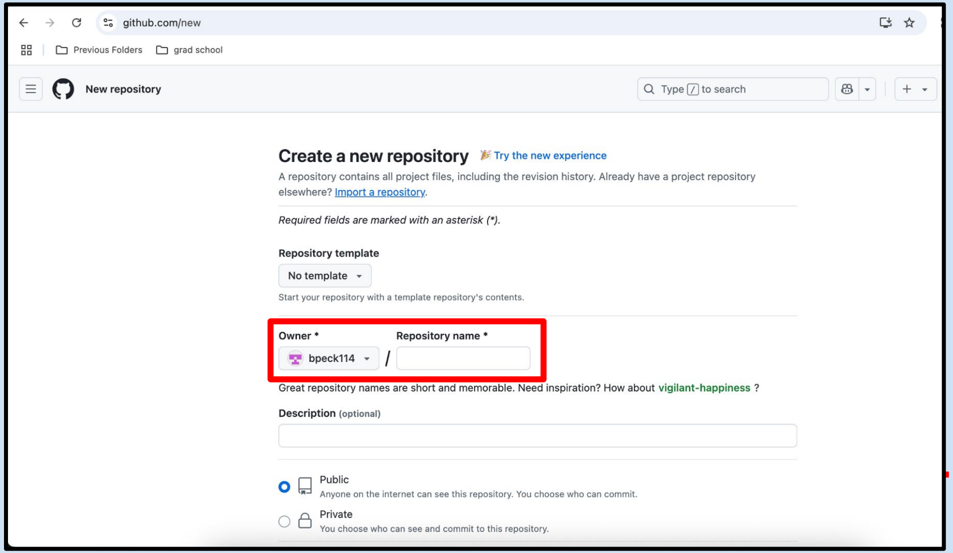
Make a new repository



A drop-down menu will appear.

Click on the tab that says: New repository

create a new repository



Another new page will open titled: <u>Create a new repository</u>

Go to the section called: <u>Repository name</u>

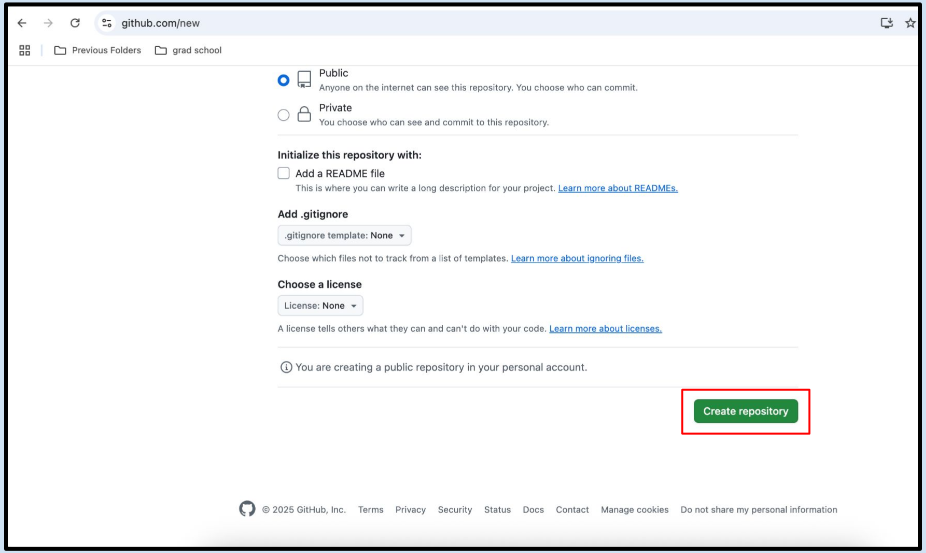
same name as local repo

Give it the same title as what you called the repository on your computer!!

So: yourname

In my case, my name is Brianna. I called my repository brianna

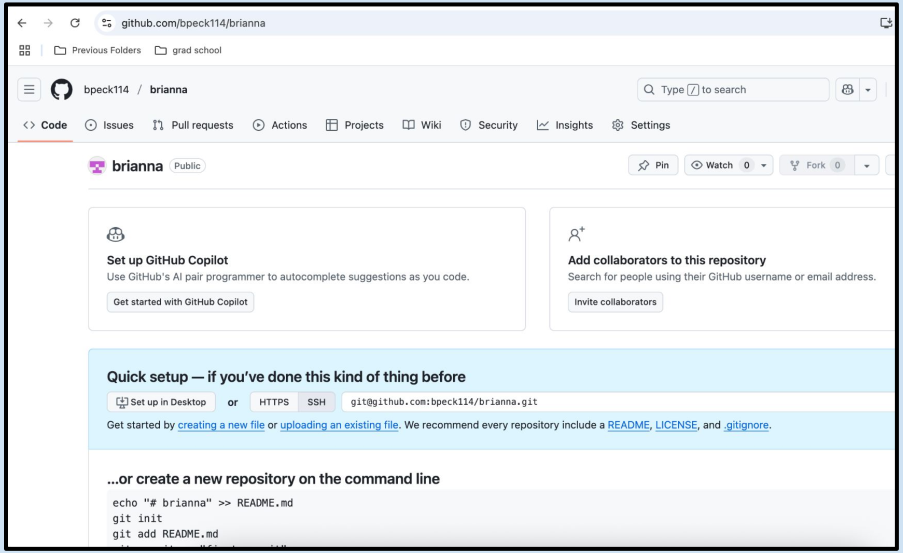
create the new repository



Leave all other settings on default

Scroll to the bottom of the page. Click on : Create repository

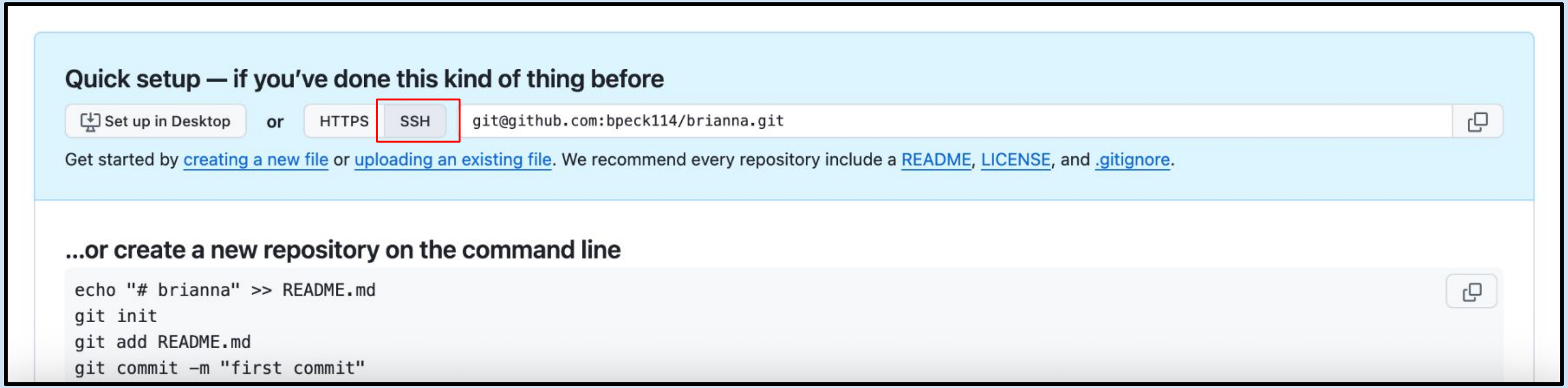
opens another new window



Another new window will open after creating your remote repository

Connect Local + Remote

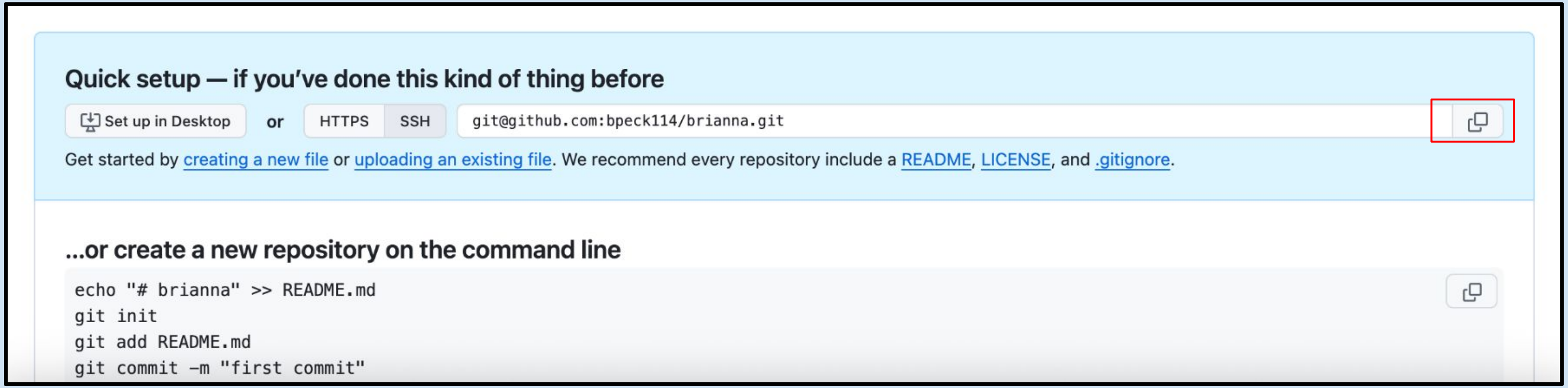
COPY THE SSH LINK, NOT HTTPS



Go to the Quick Setup section on new window

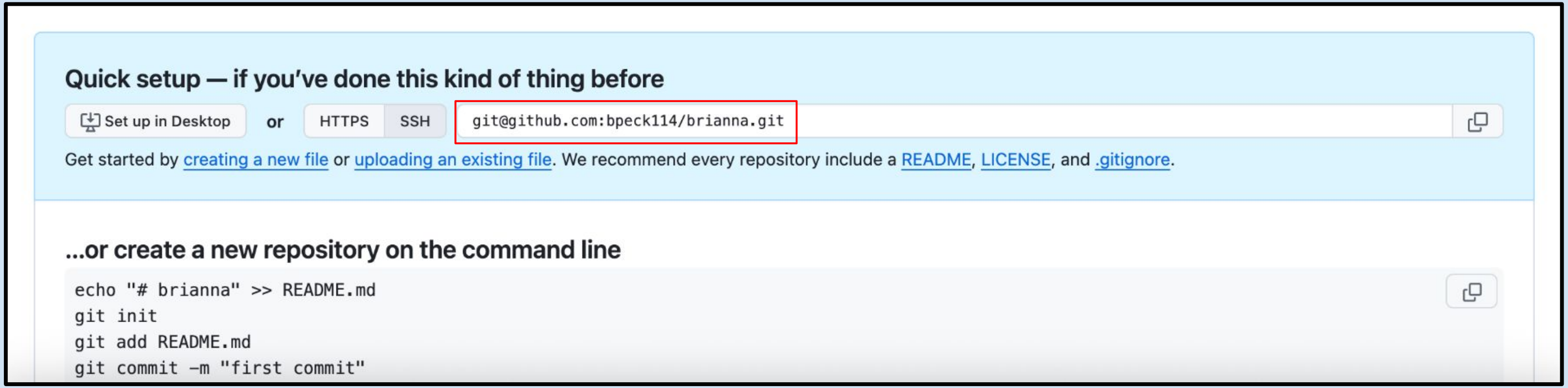
IMPORTANT!!! Make sure you have the SSH tab highlighted!

COPY THE SSH LINK, NOT HTTPS



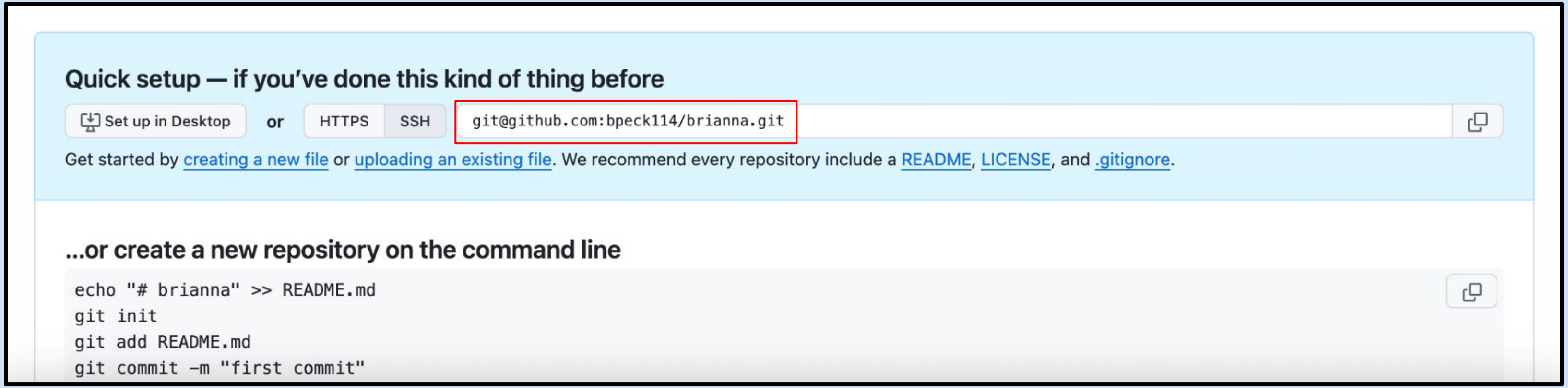
Click on the clipboard icon on the right-hand side of the screen to copy the link

Copy the SSH link, not HTTPS



It happens every year, but there is always someone who copies the wrong link, make sure to copy the SSH link!

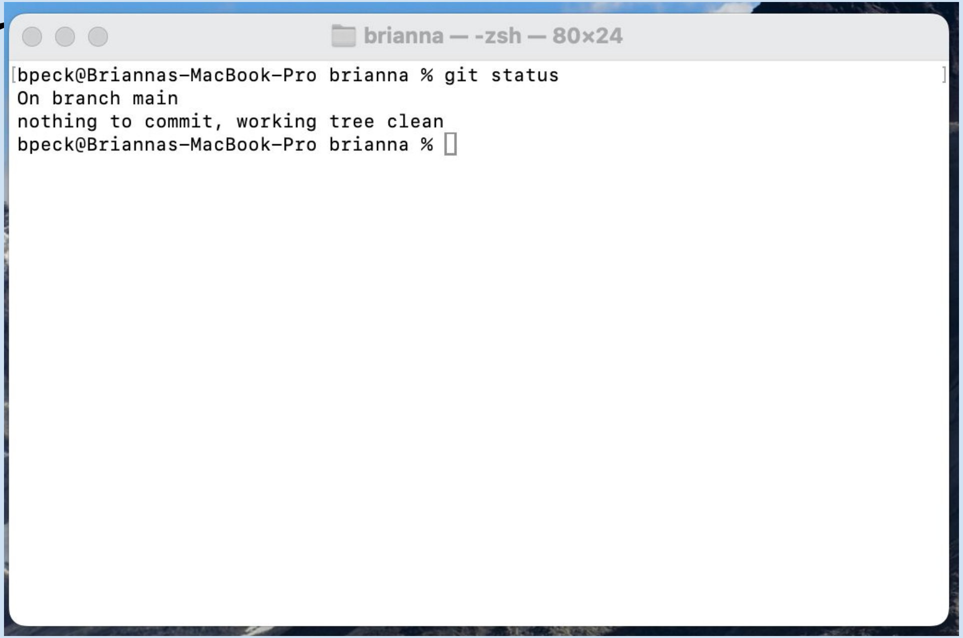
COPY THE SSH LINK, NOT HTTPS



The SSH link ends with .git,

- If the link you are copying DOES NOT end with .git then you are copying the WRONG one.

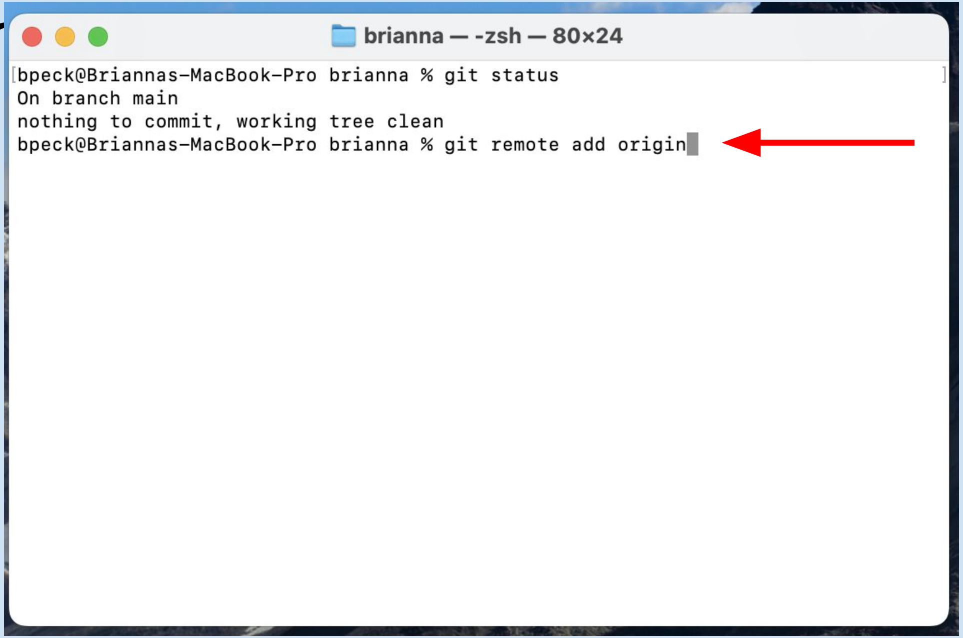
Open the Terminal



Open up the terminal again

Make sure you are in the yourname repository, otherwise this next step won't work!

ADD THE REMOTE ORIGIN

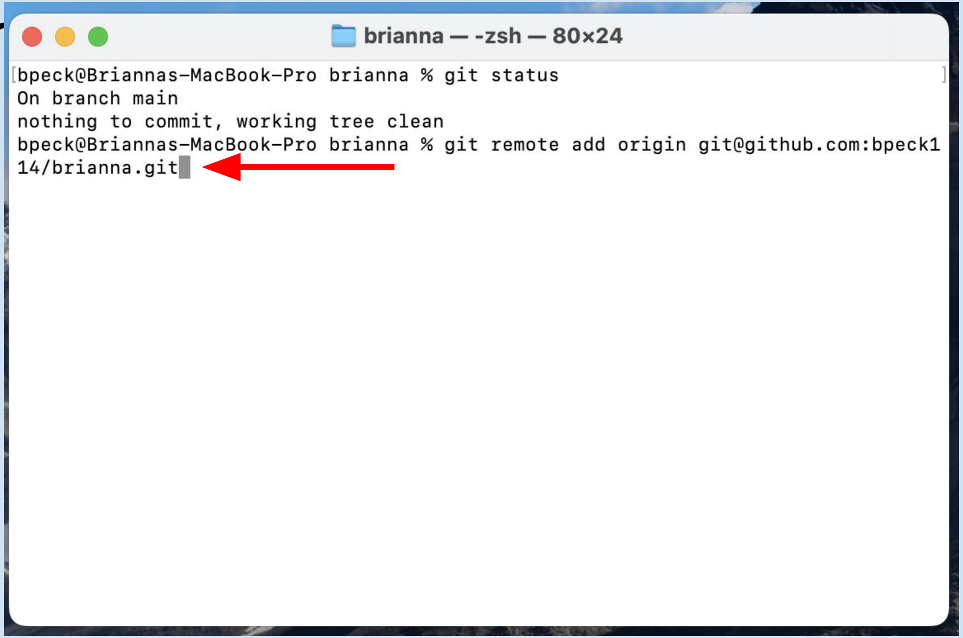


<u>DON'T Hit Return/Enter yet,</u> we will work through this command in steps

Type:

<u>git remote add origin</u>

Paste the link you copied

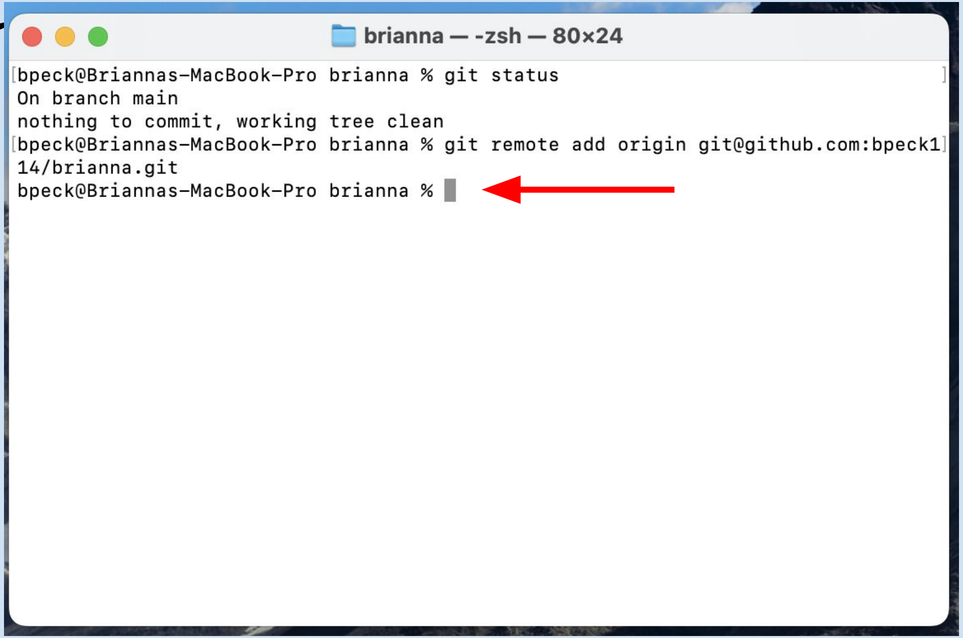


After those lines, paste the url you copied from GitHub

Make sure it is the SSH url!

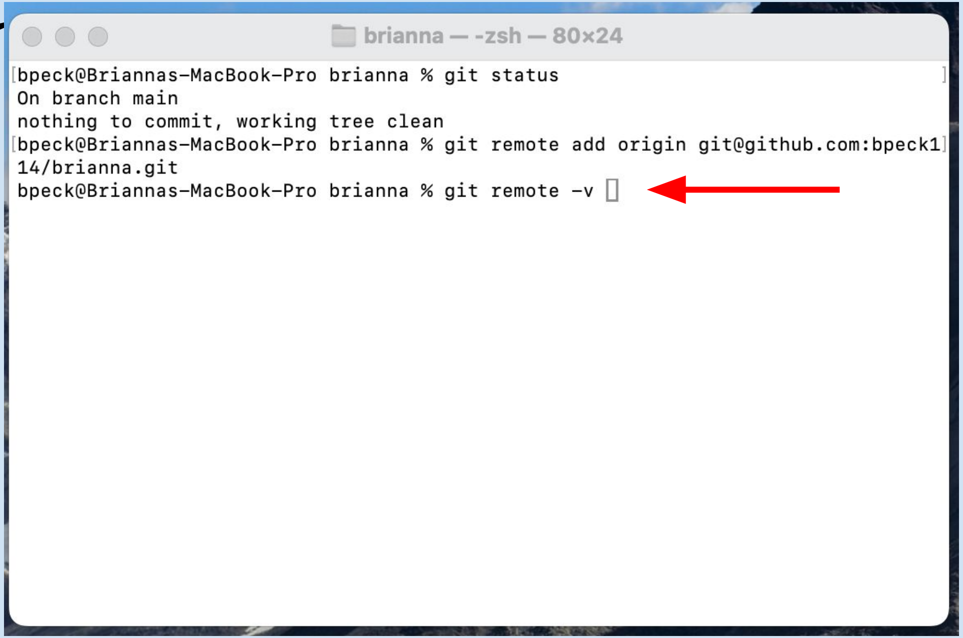
Please, please, please double check or you will get an error

Paste the link you copied



Nothing will show up afterwards, but let’s check to make sure the connection is established

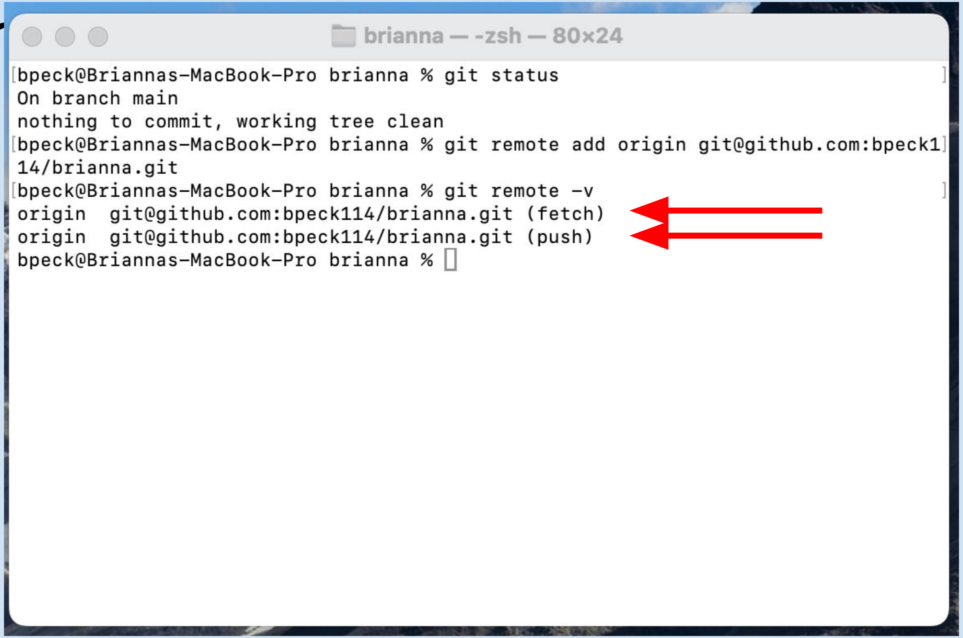
Check the Connection



Type: git remote -v

This command will get check the connection between our local and remote repositories

CHECK THE CONNECTION



origin = another word for remote repository

Then we see the link we pasted into our command line

fetch = the remote repository where save branches

push = the remote repository where save commits

CHECK THE CONNECTION

brianna — -zsh — 80×24

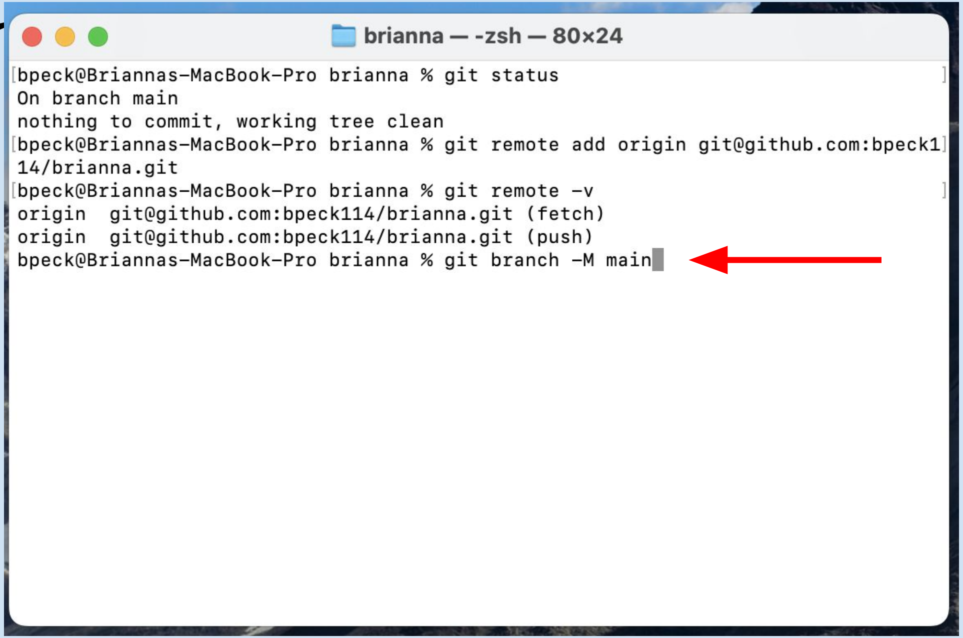
```
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck1
14/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna % 
```

Our local and remote repositories are now in communication

But no information has been sent between them

There are a couple more steps

set remote branch as main



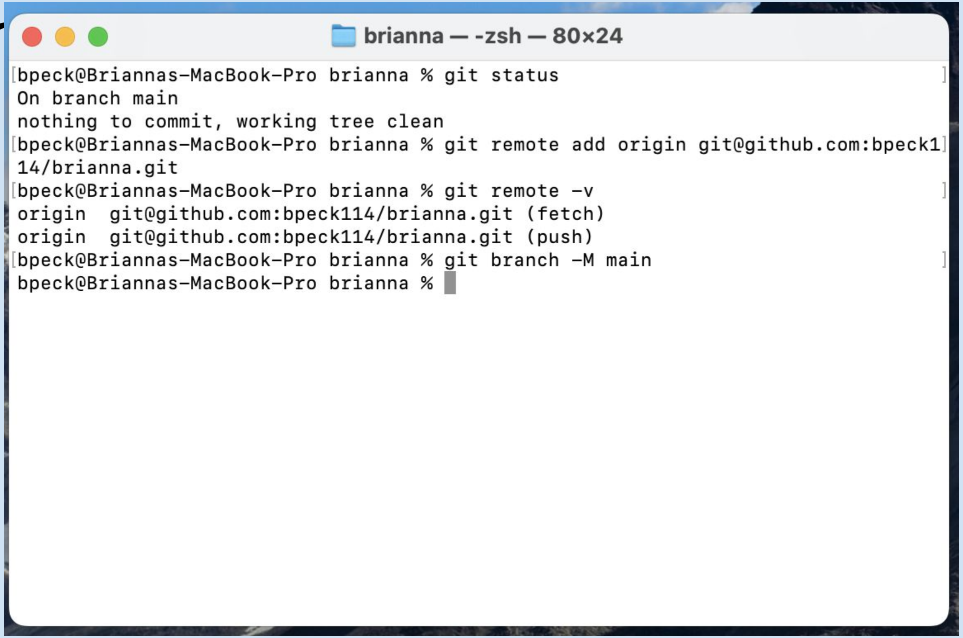
To make sure that we are pushing to the proper branch

Call:

git branch -M main

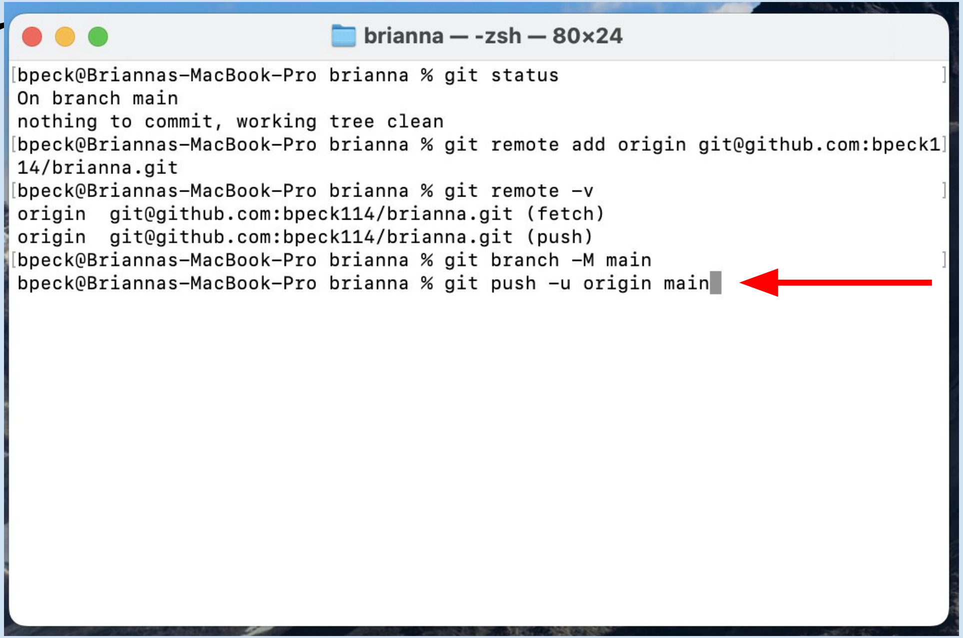
You don't need to know what branches are yet, just call the command

set remote branch as main



The last step is to officially save our changes for the first time!

PUSH TO THE MAIN BRANCH

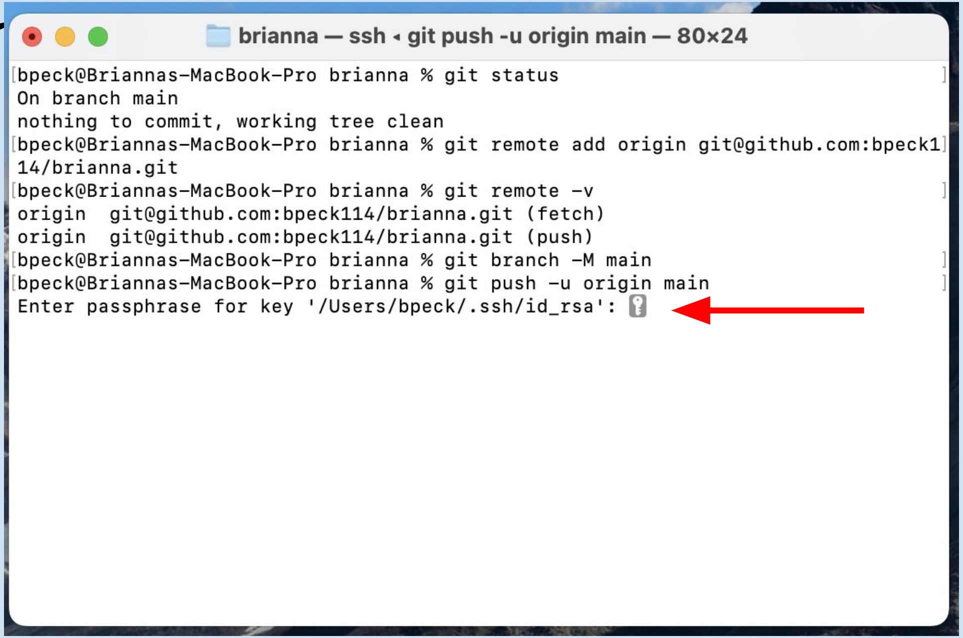


Type:

git push -u origin main

Now you are officially telling your local repository to save changes to your remote repository

Enter your password

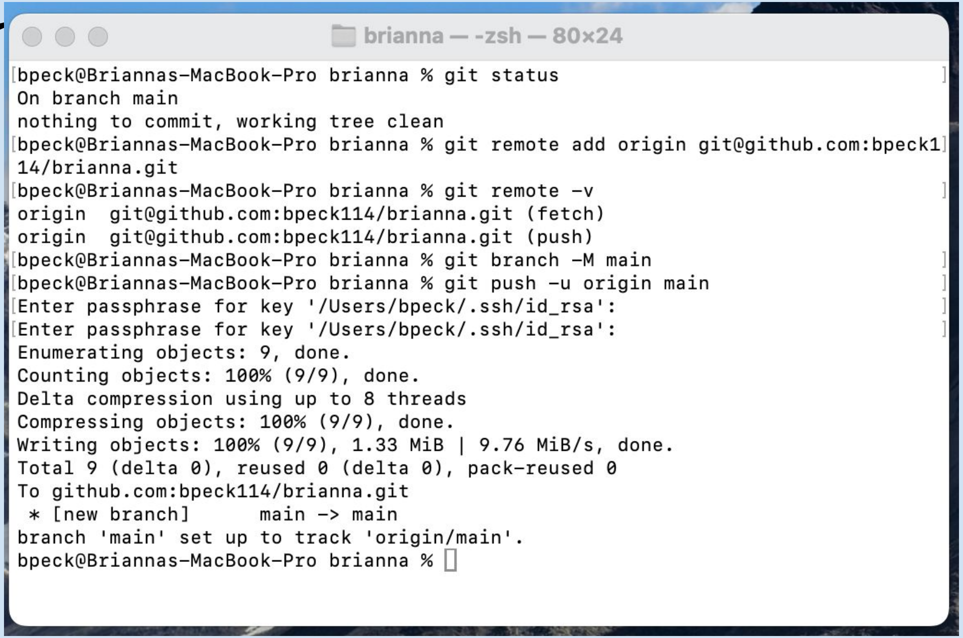


You will be asked to type in your SSH password

You created this while following the Installation Guide

Nothing will show up for your privacy!

save your changes remotely

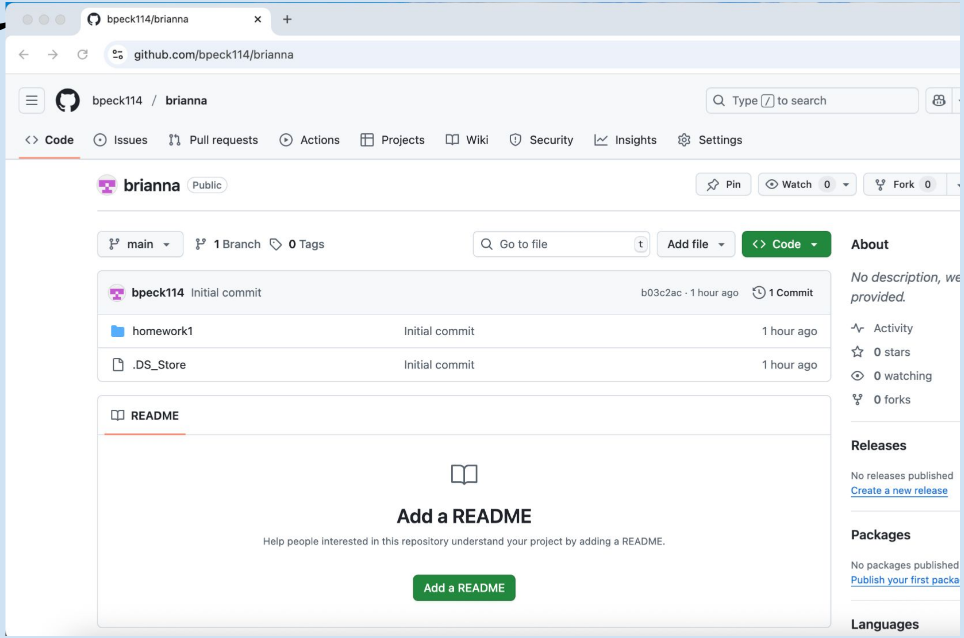


```
brianna — -zsh — 80×24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck1
14/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna % git branch -M main
bpeck@Briannas-MacBook-Pro brianna % git push -u origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.33 MiB | 9.76 MiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:bpeck114/brianna.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
bpeck@Briannas-MacBook-Pro brianna % 
```

After typing in your password you will get this really long message

But as long as your don't see <u>error</u> in the words, you are good to go!

save your changes remotely



If you go back to your web browser with GitHub

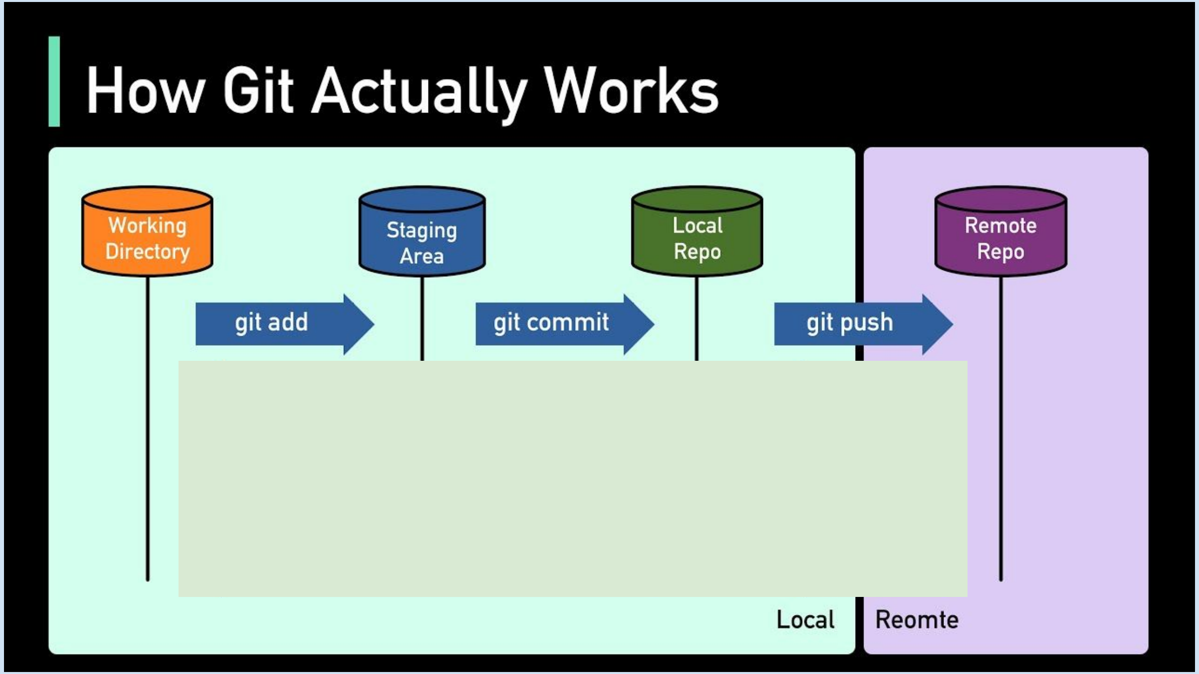
You should now see your homework1/ folder!

04

Key Takeaways

Most important information

A Helpful Diagram



*more of this diagram will be revealed later

Background Information

3 key commands for saving your work with Git/GitHub:

- *git add .* OR *git add <file>*
- *git commit -m "message"*
- *git push origin main*

Follow those three steps anytime you want to save changes!

Background Information

Oh no! I ran into a problem, how do I fix it?

Call these commands to learn more about the issue:

- *git status*
- *git log* OR *git log -1*

Use GitHub for Homework

From now on, you will be turning in your homework via GitHub repositories on Gradescope

Homework #2 will walk you through how to do so

If you need help, please see the reference slides at the bottom of this slideshow

Let's practice

Practice Question

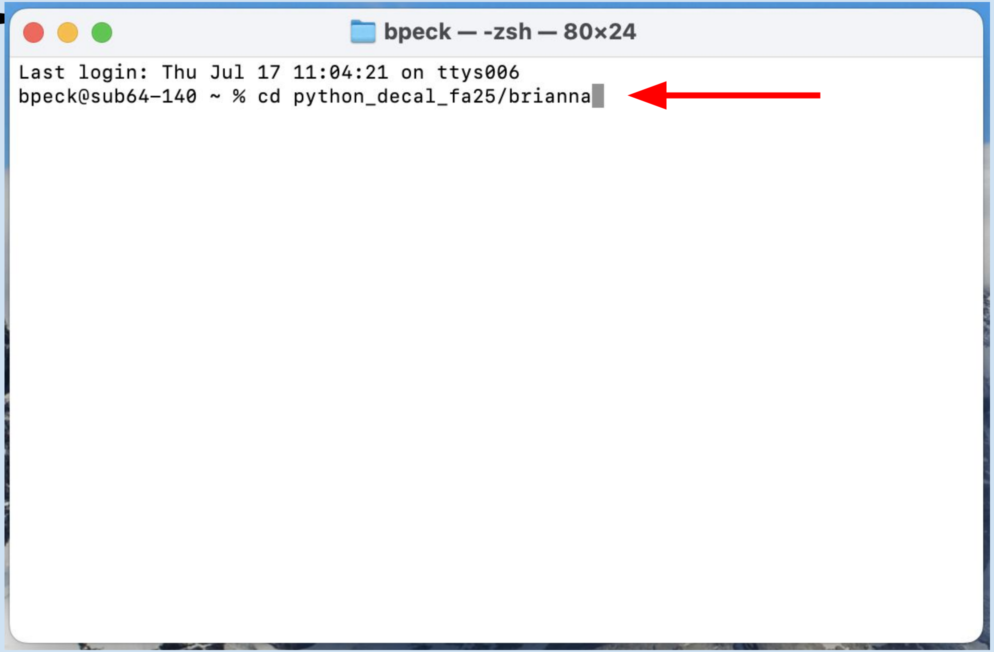
Think by yourself and then discuss these questions with a partner:

- Then we will go through it together as a class

Let's say you just opened the terminal, what commands would you need to call to do the following:

- Enter the yourname directory
- Create a new file called test.py and
    - Code: print(5 + 5)
- Run the Python code on the terminal
- Save your changes to GitHub
    - *Hint:* git add, git commit, git push

Answer: Change Directories

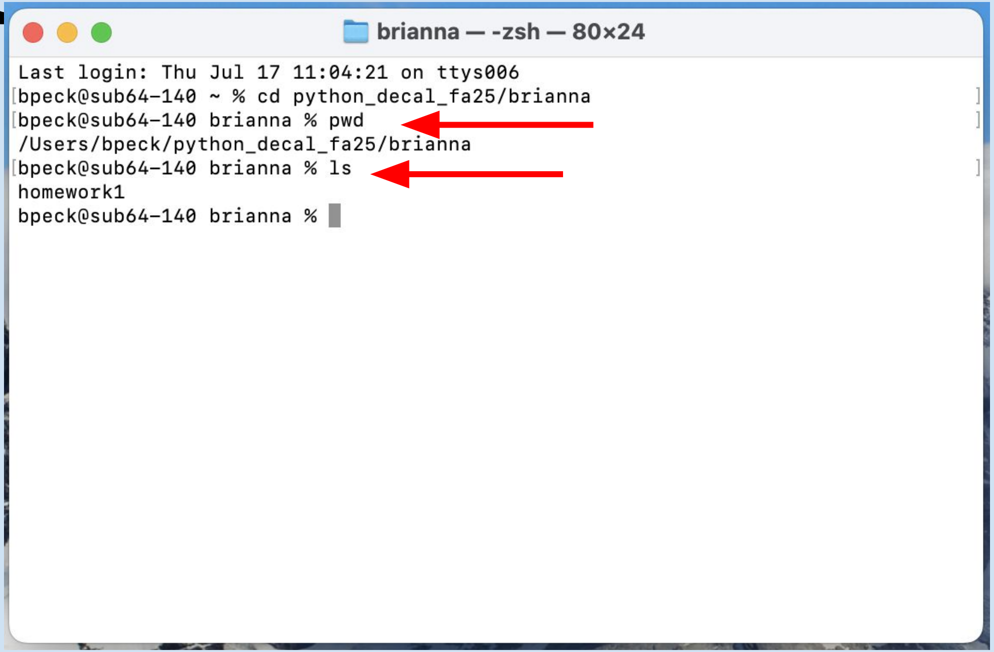


Start by opening up the terminal

Then write the file path to your yourname directory

Use the cd command to change directories

Answer: call pwd and ls



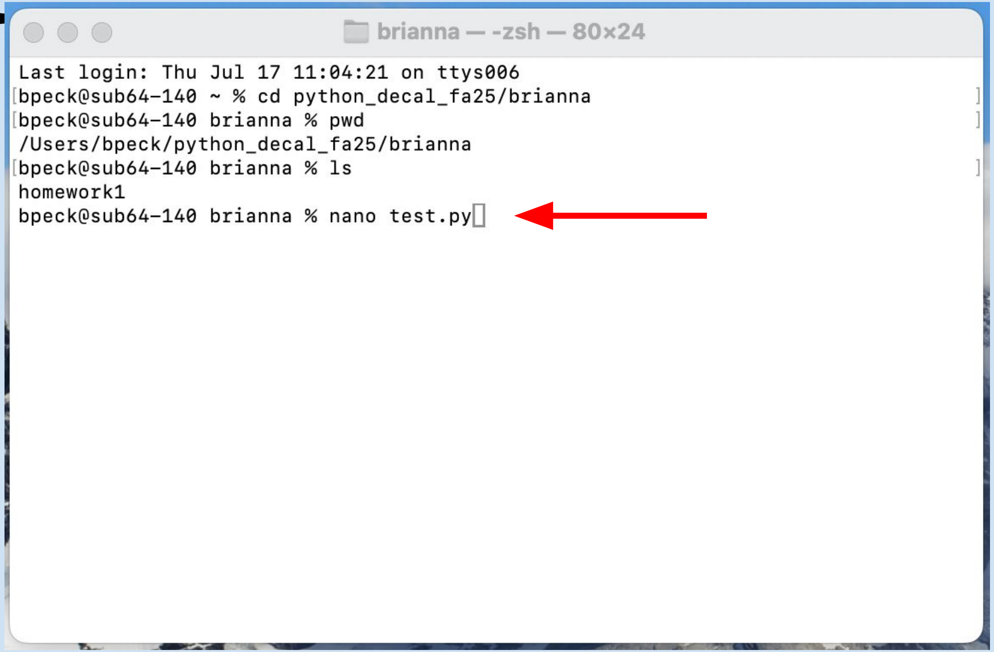
To double check we are in the right location, it is always good to call:

pwd

ls

To figure out your file path and the contents of your directory

Answer: use nano to edit

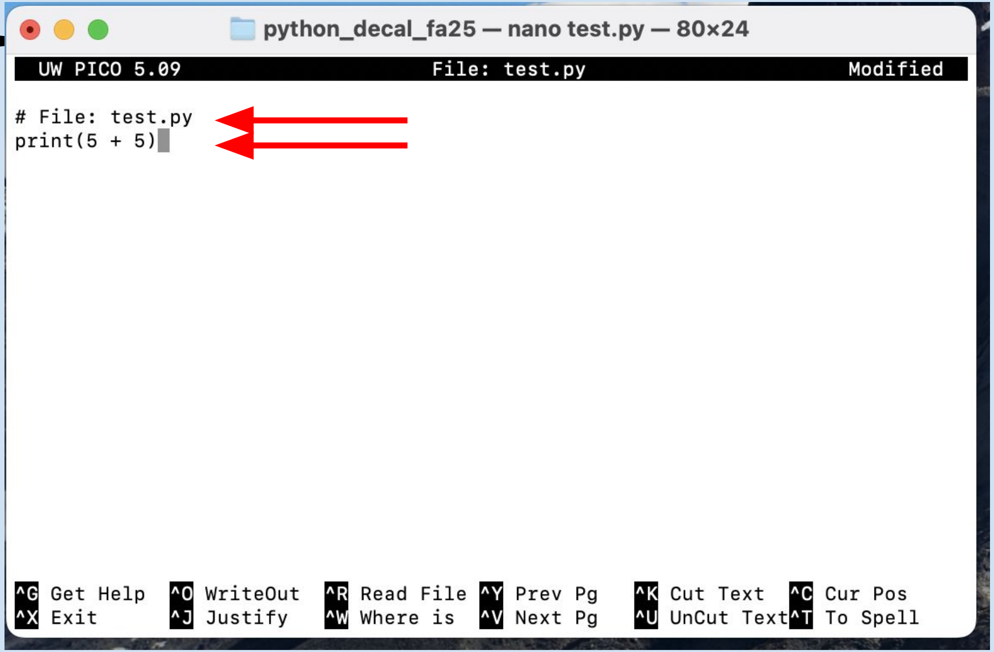


We can immediately open a file by calling:

nano test.py

But you can also use the touch command if you don't want to edit the file

EDIT THE FILE WITH nano



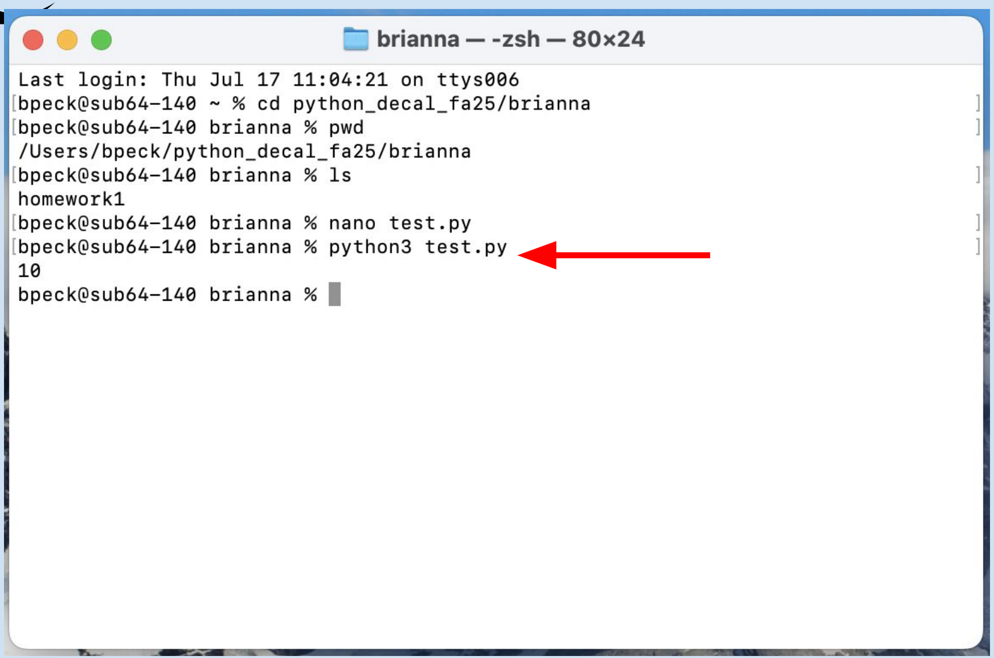
It's always good to add a comment at the top of the file indicating it's <u>file name</u>

Then type:

<u>print(5 + 5)</u>

Save your changes and close <u>nano</u>

Answer: Run Script in Python

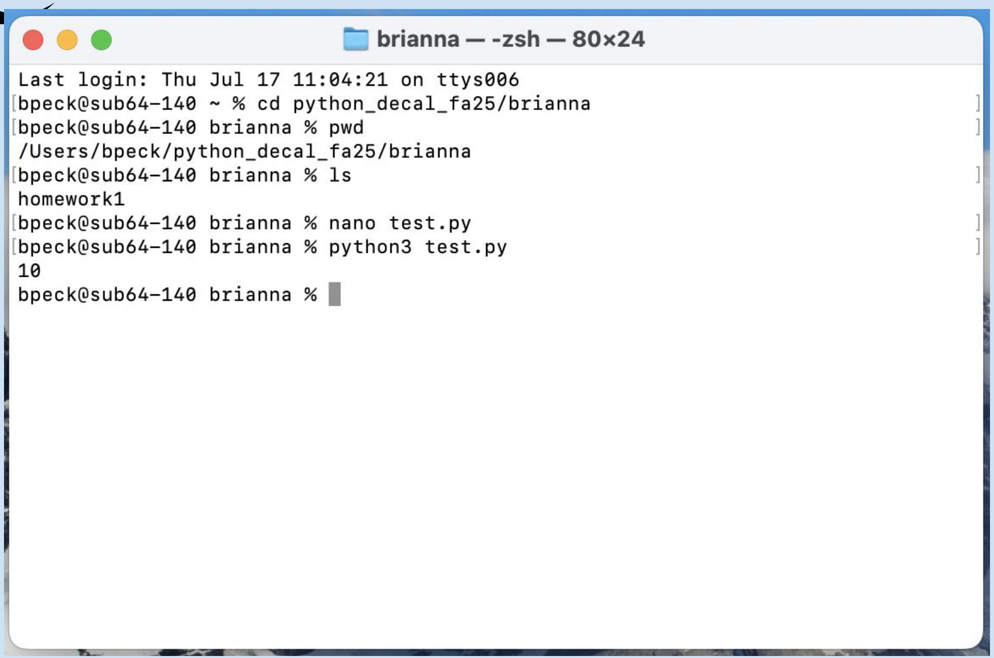


To run your script directly on the terminal, you can call:

<u>python3 test.py</u>

Then your terminal will return the operation

Answer: Run Script in Python

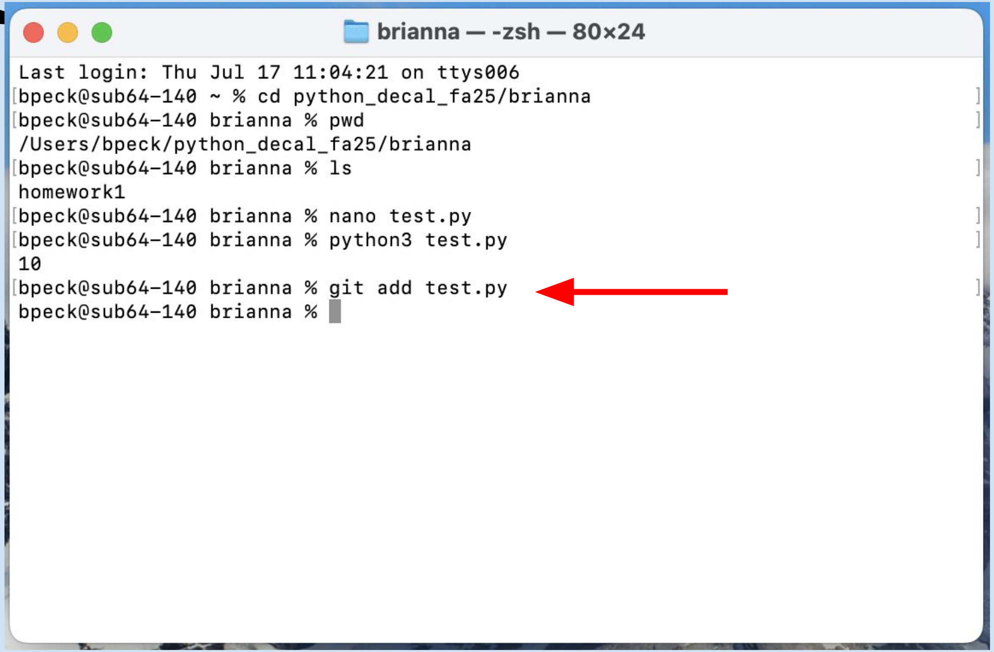


To save our changes with git, we need to follow three steps:

- git add
- git commit
- git push

With needed modifications, of course

Answer: ADD Changes Area



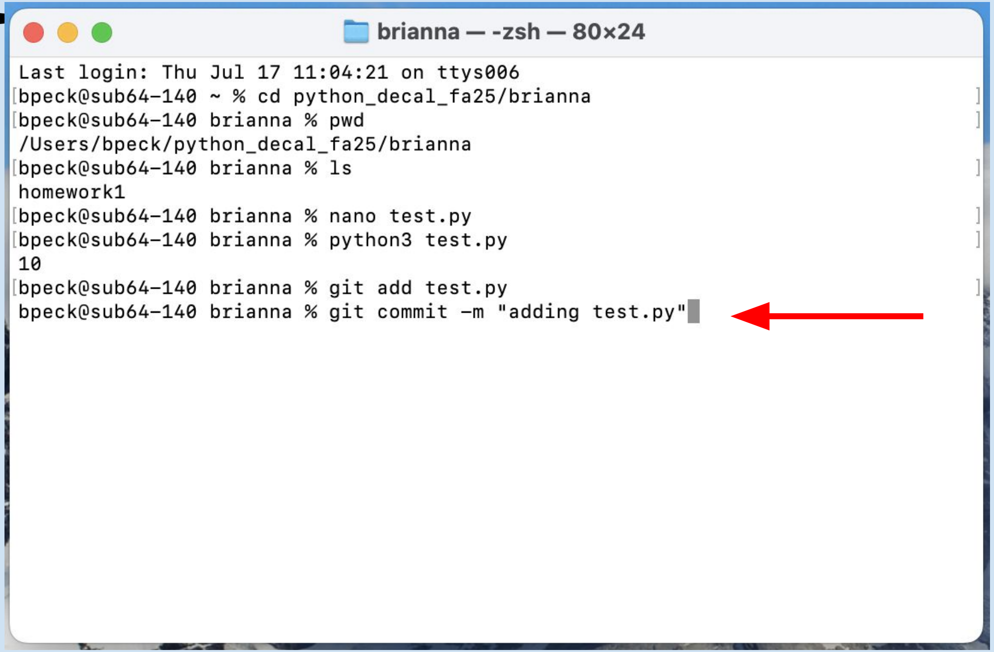
First up is <u>git add</u>

Let’s practice only adding the file we created

Type:

<u>git add test.py</u>

Answer: save Locally



To save our changes locally, we need to use <u>git commit</u>

This moves our changes from the <u>staging area</u> to <u>version history</u>

Type:

<u>git commit -m “adding test.py”</u>

Answer: save locally

```
brianna — -zsh — 80×24
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd brianna
bpeck@Briannas-MacBook-Pro brianna % nano test.py
bpeck@Briannas-MacBook-Pro brianna % python3 test.py
10
bpeck@Briannas-MacBook-Pro brianna % ls
homework1       test.py
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
 1 file changed, 2 insertions(+)
 create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna %
```

The last step is to <u>push</u> or saved history from our local computer to our remote repository

Answer: save remotely

brianna — -zsh — 80×24

```
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd brianna
bpeck@Briannas-MacBook-Pro brianna % nano test.py
bpeck@Briannas-MacBook-Pro brianna % python3 test.py
10
bpeck@Briannas-MacBook-Pro brianna % ls
homework1       test.py
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
 1 file changed, 2 insertions(+)
 create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
```

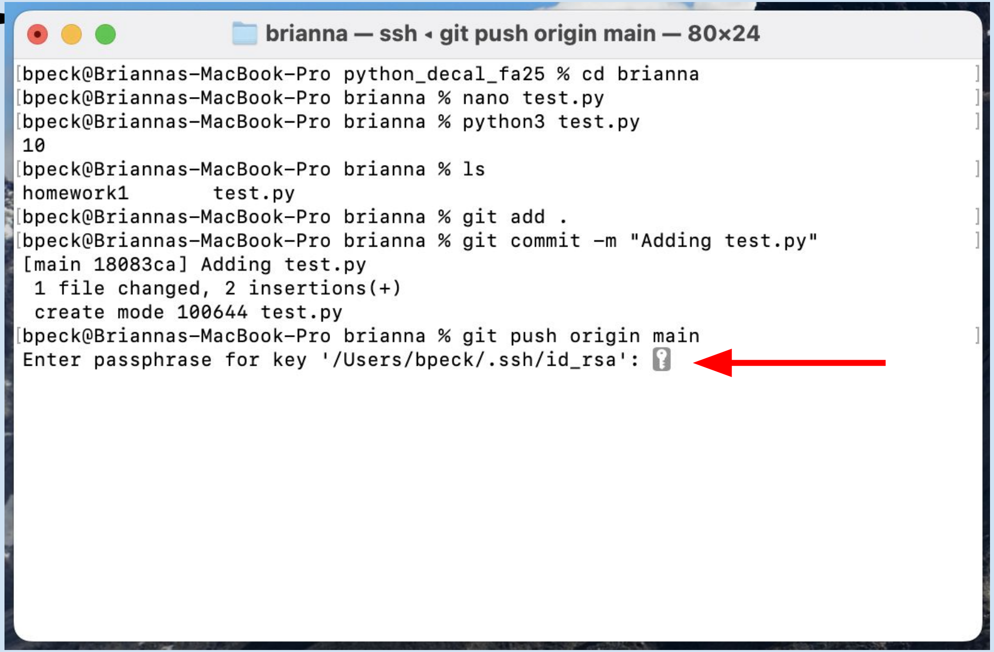
When you call git push, make sure you are pushing to the correct branch

Which for us is main

So call:

git push origin main

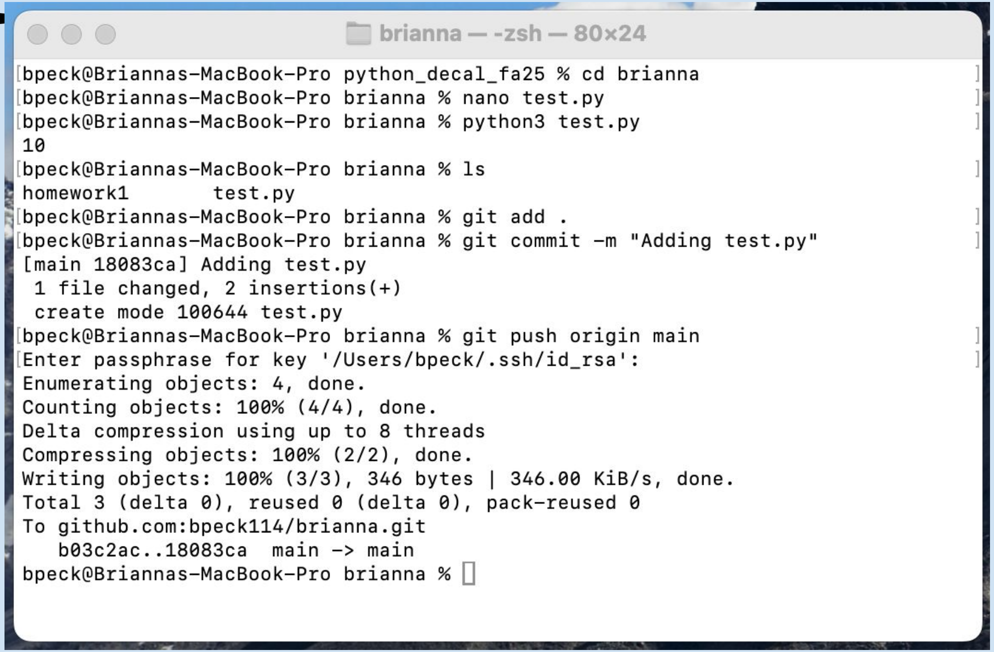
Answer: Enter Password



You will be asked to type in your password again

Hidden for your privacy

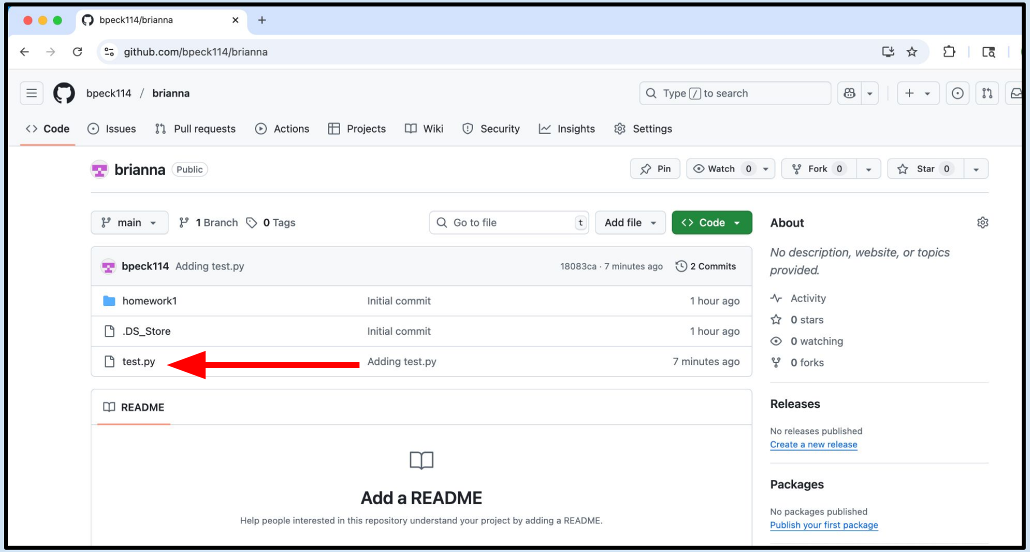
Answer: confirm no errors



```
brianna — -zsh — 80×24
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd brianna
bpeck@Briannas-MacBook-Pro brianna % nano test.py
bpeck@Briannas-MacBook-Pro brianna % python3 test.py
10
bpeck@Briannas-MacBook-Pro brianna % ls
homework1       test.py
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
 1 file changed, 2 insertions(+)
 create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 346 bytes | 346.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:bpeck114/brianna.git
   b03c2ac..18083ca  main -> main
bpeck@Briannas-MacBook-Pro brianna %
```

If no error message appears then you have properly saved your changes

Answer: Check on GitHub



When you refresh GitHub, you should now see test.py!

05

Lecture Engagement

Show us what you know!

Lecture Engagement

Fill out Lecture Check 3!

Always due two days after class.

Takes <5 minutes to complete.